Mrunal's Economy for UPSC Prelims-2026-27 (PCB16-Powerhouse)

Pillar1A1: Money: Barter to Bitcoins

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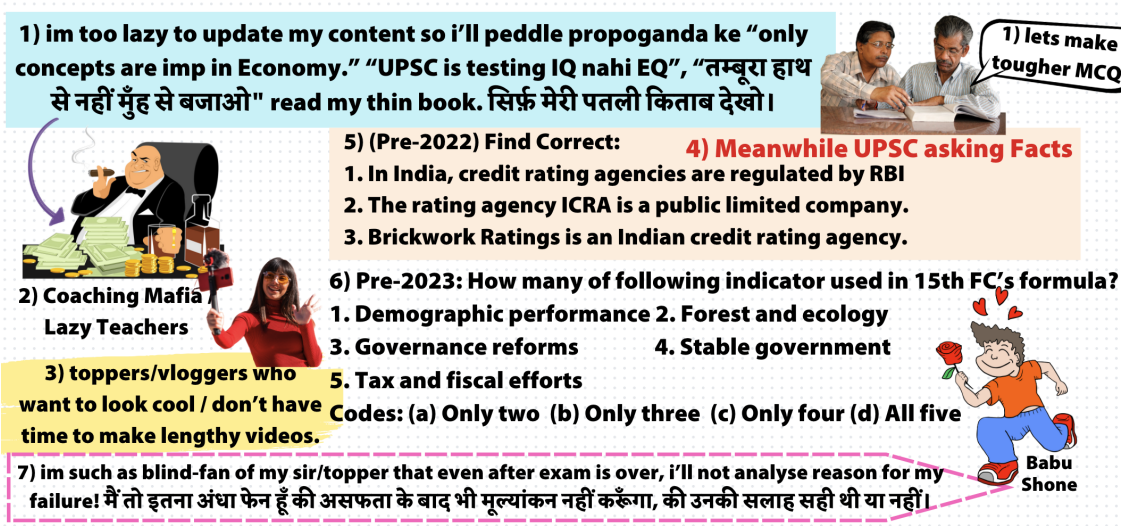
10 Brief Info about Economy course (PCB16-Powerhouse)

Economy Area	Covered?	Comment, if any
Prelims	✅ Yes	Yes- theory, current, NCERT, Budget, Economic Survey, Govt schemes, MCQ, PYQ- everything covered. Course Page Link
Mains	✋🚨No	Separate course for it. You've Two Options: (A) Wait for next Live Mains Course (QEP9) in Aug'26 **OR** (B) finish from Past Recorded Course (QEP8)
Micro Economics	✋🚨 No	UPSC asked only 2 question in last 15 years. Finish from Recorded Course of 2023-May. Click Me

- **Syllabus, Strategy, Booklist, Cut-off:** https://mrunal.org/syllabus
- **2026 strategy / roadmap / planner video**: Watch Lecture#1 of PCB15 Course.
- **Economy Booklist = Explained in LECTURE Number#2 or 3.**
- Basically this PCB-**Raftaar** Course deals only with **Prelims**-Specific **Economy**.
- **Old Students (PCB1to15) Need NOT Sit again. They shld see FREE UPDATEs [Win##] at** Mrunal.org/win26 as per the table given in it.

10.2 Myths about Economy for UPSC Prelims



I'm not making Memes for everything, otherwise pages will increase so just accept following:

Myths	Reality
In Economy, ONLY concepts are important for UPSC.	Both facts and concepts are asked.
if something is not asked in the last 3 years, then such topic	UPSC examiner asks trend-

Myths	Reality
will never be asked.	breaking questions.
if some economy topic is older than one year, then it is not important. Only last one year current affairs is important.	UPSC asked many Economy MCQs from events older than one year.
We've to read economic-times, business line, etc newspapers.	NOT required.

10.3 PYQ Analysis: Increased diversity in Economy topics in Prelims

- From table given below, you can see that the diversity and variety of economy MCQs questions has increased/changed in recent Prelims, so it is inevitable that handouts will get lengthy. जब प्रश्नों की विविधता और जटिलता बढ़ रही है तो ये लाजमी है कि ज्यादा नोट्स लम्बे बनाने पड़ेंगे.
- So, please do not be fooled by the "**negative-PR toolkit**" circulated by Delhi-offline coaching mafia against my notes via timepass vloggers, bikaau-royal-entry toppers and Reddit Trolls.
- Black/GREY background cell = NOTHING asked from said topic in that year.

Pillar	2026	2025	2024	2023	2022	2021	2020
1A1_Digital Payment & Bitcoin	1) Blockchain 2) UPI vs CBDC 3) Tokenization	(1) NEFT-RTGS pe fees (2) UPI deal with foreign nations	e₹ by RBI	CBDC & SWIFT	NFT ke features	NOTHING asked	Blockchain Technology
1A2_Monetary policy	NOTHING			Central Banks globally hiking loan interest rates	RBI responsible to control inflation	(1) Money multiplier increases when (2) Indian Bond Yield influenced by what factors	(1)expansionist monetary policy (2) multiplier If you withdraw 1lakh in cash
1B1_Classification Bank NBFC	NBFC Features		Foreign Banks		Credit rating agencies	Lender of last resort means what UCBs regulation	District Central Cooperative Banks (DCCBs)
1B2_NPA/ Bad Loan Burning Issues		RBI profit from seigniorage	Syndicated lending		BBB	Appointment of RBI Governor	Interest Coverage Ratio
1C_Sharemarket, companies act, CSR	1. Sustainability bond 2. M1xchange MSME 3. Crowdfunding	(1) SEBI advisory on option trading (2) AIF (3) bond holder ke rights (4) SEBI- BRSR	(1) US treasury bonds (2) C-Bond participants (3) CSR (4) ETF (5) CBLO (6) Debt-platform	InVITS pe Income Tax & SARFAESI Act (2) Capital markets ke sub-markets (3) Beta in Sharemarket (4) intangible investments (5) Capital Expenditures of	IIB pe Income Tax Convertible Bonds	Stock Market: NDS-OM & CDSL	Call Money & other instruments (2) non-financial debt

Pillar	2026	2025	2024	2023	2022	2021	2020
				a company			
1D1_Insurance	1) Aviation Insurance 2) Malhotra committee						cyber insurance
1D2_Pension			PM-Shram Yogi			Causal Workers get EPFO or not	
1D3_ Financial inclusion	FI-Index RBI			SHG ko Loans			Kisan Credit Card scheme
2AB_Tax_FC		(1) Agri pe income tax (2) FC grants (3) ED-ministry pairs		15th FC Horizontal Formula	Vodafone indirect transfer	Black Money is bad why?	
2D_FRBM Deficit Disinvestment	Crowding Out Effect	(1) FD calc. (2) PD calc. (3) budget classfn.			G-Sec and Household Savings		FRBM Docx
3A_BoP import export FDI/FPI			Apple import	Switzerland Gold export (2) India ka global export data & PLI	Tea Board ka Dubai Office	FDI types of instruments	FDI Meaning (2) India import-export data (3) SriLanka Trade (4) WTI Oil Contract
3B_ currency exchange	Hilton-Young Commission			RBI s Sterilization	NEER-REER currency exchange and RBI Action (2) ECB & Fed Tapering	Devaluation	Gold Tranche (2) GFC- what NOT to do
3C_intl.org agreements		IBRD loans		USA-EU Trade and technology Council (2) EU s Stability Growth Pact	IMF Loan facility (2) G20 Common Framework (3) AIIB-MTCR-SCO		TRIMS Agreement
4A_AGRO	Rainfed Area Development	Gokul Mission	(1) farmer capital (2) Land records (3) Grains council	Small Farmer Large Field (2) Nigerseed pe MSP			price of rice affected by public investment in agri MSP All/ NEVER
4B_Mfg Services E-Commerce MSME IPR	1. Oeko-Tex Eri Silk 2. Dropshipping 3. ONDC means	(1) circular eco (2) Coir board	Sectors classification	MSME & PSL	Videshi e-commerce firms		

Pillar	2026	2025	2024	2023	2022	2021	2020
4C_NITI PC FYP; unemployment			Venezuela Economic model		Vietnam economy growth (2) Org 4 industry dispute		1991 liberalization impact on rural area
4D_GDP					real sector in the economy	Counter-cyclic policy to combat recession	
4E_Inflation						Demand-Pull inflation (2) Printing money causes inflation	CPI vs WPI
5A_Infra-Energy Water	REE Minerals	(1) PM Surya ghar (2) critical minerals (3) PNGRB			Coal Controller s Organization (CCO)		
5B_Infra-Transport	1. Vizhinjam Seaport 2. Sagarmala 3. India ASEAN Connectivity		Greenfield-Airports	Private Port etc Infra Corridors			
5C_Infra-Rural Urban	Gram Swaraj Scheme			S3i ke Infra sectors	Sanitation Coalition		MPLADS
6_HRD-Health Edu, Poverty, SDG	1) PH scheme 2) MPI Index		(1) TFR (2) PM-Surakshit Matritva (3) nations ↓ pop.	(1) Janani Suraksha (2) Anaemia Mukt Bharat	Ayushman Bharat Digital Mission		
7_Microeconomics						substitute & inferior goods	
TOTAL	23 MCQs##	20 MCQs	20 MCQs	20+ MCQ##	20+ MCQ	14 MCQ	20+

- **Free Download Topicwise PYQ booklet at https://unacademy.com/content/upsc/downloads/**
- **## Note on above table**: Every teacher always wants to show that more number of questions are asked from his subjects. E.g. Economy teacher will say WTO = economy topic whereas IR teacher may disagree. No need to get into 'emotional-fanboy-outrage' around it.

10.3.1 PYQ Analysis: Marks Distribution in UPSC Prelims and Mains

6 Pillars of economy	Prelims P1 (GS 100 MCQs)									Mains GSM-1-2-3 (750 Marks)								Utility in exam
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2018	2019	2020	2021	2022	2023*	2024	2025	
1_Money Banking, Finance Insurance	7	7	9	6	8	7	10	7	10	0	0	15	15	0	0	0	0	Most useful in Prelims but less in Mains
2_Budget, FC, GST	4	1	1	1	3	1	0	6	1	25	25	15	20	*0*	*15*	*0*	*30*	*Second lowest utility after 5_Infra but Pre'25 was trend breaker*

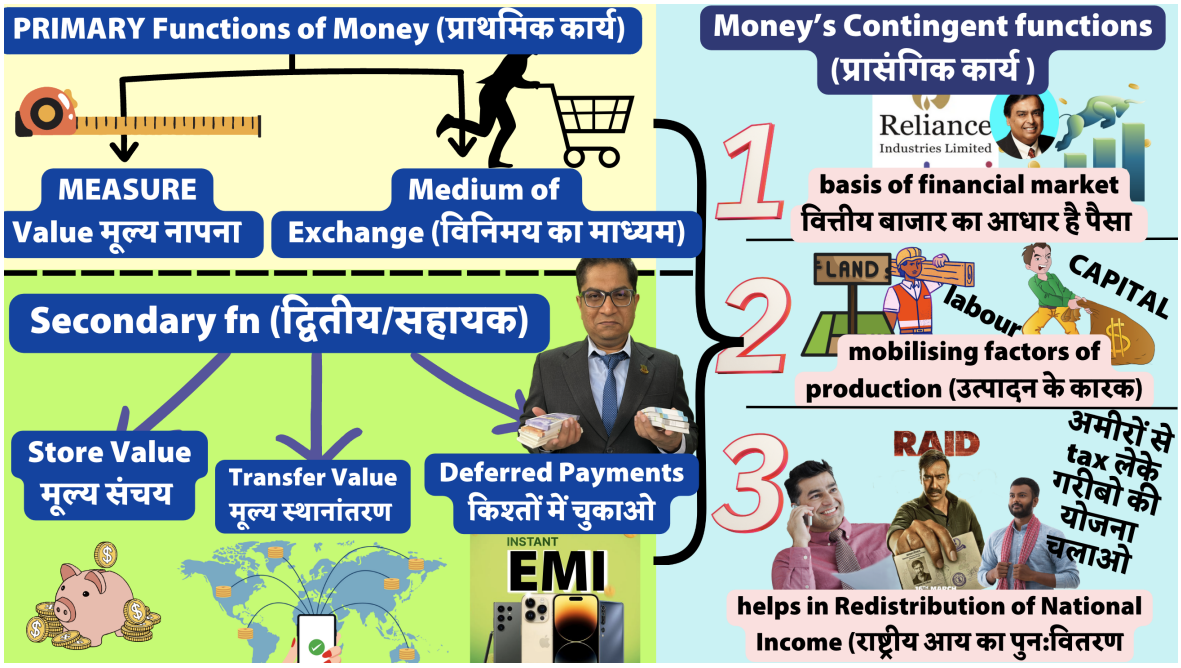
3_Intl_Trade	4	7	7	2	4	3	1	1	1	30	0	15	15	0	0	0	10	*Third lowest utility after 2_Budget.*
4_Sectors, GDP, Inflation	3	8	5	5	4	4	5	3	4	50	65	55	60	85	110	60	90	Useful for both prelims and Mains.
5_Infra	1	3	1	0	2	4	1	3	5	15	15	15	30	*20*	*10*	35	20	*Lowest utility among 6 pillars. But pre'25 trend breaker*
6_HRD	6	2	0	0	0	2	3	0	2	60	75	85	45	20	95	45	60	More useful in Mains than Pre
Total	25	28 Q	23	14 Q	21 Q	21	20 Q	20 Q	23 Q	180 m	180 m	200 m	185 m	125 m	230 m	140m	210m	---

10.4 1A1: Barter System & its disadvantages: (वस्तु विनिमय प्रणाली)

Barter system was introduced by **Mesopotamian** tribes. It has following challenges: (चुनौतिया)

- Double co-incidence of Wants (आवश्यकताओ का दोहरा संयोग नहीं होगा तो व्यापार असंभव).
- Search Cost & Transaction cost is high. (सही दाम पर चीज ढूँढने व लेनदेन की लागत बढ़ जाती है)
- Storage of perishable commodities is difficult, results in loss of value. (नाशवान वस्तुओ का मूल्यहास)
- Doesn't encourage specialization and division of labour (श्रम विभाजन को प्रोत्साहित नहीं करता).

10.5 Money: Functions of (पैसे के कार्य)



10.6 Money: Evolution & Types (मुद्रा की उत्क्रांति व् प्रकार)

Commodity Money → Metallic Money → Paper /Token / Representative / Fiat Money → Bank Money, Deposit Money → Crypto Currency

10.6.1 Commodity Money / वस्तु मुद्रा (Intrinsic value: अंतर्भूत मूल्य? Yes)

⇒ Iron Nails, Bear Pelts, Cocoa Beans, Whale Teeth, Gold Nuggets

⇒ **Problems?** Perishable, not uniform, not pure, foreigners may not accept. नाशवान, एकसमान नहीं, शुद्ध नहीं, अस्वीकार भी कर सकते है.

10.6.2 Metallic Money / धातु मुद्रा (Intrinsic value? Yes)

⇒ Traders & Kings stamped their marks on gold nuggets for uniformity & trust. (सोने पर अपना चिन्ह/छाप लगा देते – ताकि लोगों को उसकी शुद्धता पे यक़ीन हो)

⇒ Indo Greek kings & **Kushana** kings issued gold coins, but Gupta Gold coins most spectacular- king is playing Veena, shooting animals, standing with wife.(इन राजाओने सोने के सिक्के निकाले थे)

⇒ Delhi Sultanate Kings: **Silver Tanka**. Sher Shah Suri **Rupiyah** silver coin. Akbar: **Muhr**.

Further SELF STUDY: History of Indian coins from History Lecture/Notes.

10.6.3 Metallic Money ⟶ Full Bodied vs Token Coins

Table 1: Types of Coins

Full Bodied Coins	Token Coins (प्रतीककात्मक)
Gupta Coin: ~8 gm GOLD	**50 paisa modern coin: ~4 gm steel**
Intrinsic Value (अंतर्भूत मूल्य) is Equal to or MORE than its Face Value (अंकित मूल्य)	LESS. (धातु एकदम सस्ती वाली इस्तेमाल करते हैं या बहुत ही कम वजन की धातु इस्तेमाल करते हैं)
Challenge? ⇒ **Debasement**: ↓ decreasing the amount of metal in coins. Usually happens when king's rule gets weak/poor ⇒ e.g. Aurangzeb, Roman Kings. (सिक्के में धातु की मात्रा को कम किया जाए तब लोगों का उस राजा के शासन से विश्वास कम होने लगता है)	⇒ **1330s:** Tughlaq's Token coin experimentation = #EPICFAIL (Why? Read in History). ⇒ Modern Indian coins are token coins. Cupronickel metal used to discourage melting. ⇒ **Coinage Act 2011** prohibits melting of coins. (सिक्का पिघलाना एक अपराध है)

FAQ by Zhande-walle Babushone: I've 25 jaat ke Qs about whether "X" can be called "Y" & Trump's decision to cancel Penny coin. Ans. As Coaching teacher, I have finished the topic as per its exam-utility. Further Ph.D doubts Ask chatGPT.

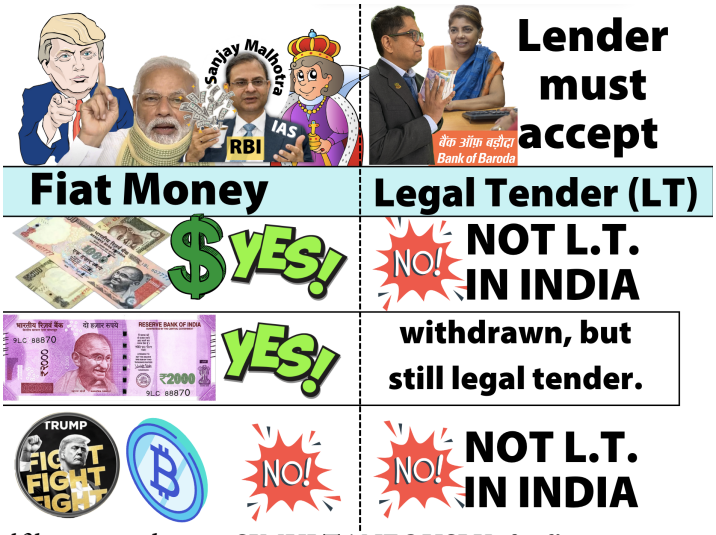
- Penny (1 cent) stopped in USA (2025) 1 dollar = 100 cents. A penny = 1 cent.
- It costs nearly 4 cents to make 1 penny.
- 2025: US President Donald Trump has ordered the stop of minting new pennies, but existing ones will remain in circulation.

10.6.4 Intrinsic value of a coin vs paper currency note? (अंतर्भूत मूल्य)

PRECIOUS METAL COINS (धातु के सिक्के)	PAPER CURRENCY (काग़ज़ी मुद्रा/नोट)
Metal has **INTRINSIC VALUE** i.e. Metallic money =Use it as "money", or you can melt the	You can't use currency-note's paper as a notebook or diary.

PRECIOUS METAL COINS (धातु के सिक्के)	PAPER CURRENCY (काग़ज़ी मुद्रा/नोट)
metal ⟶ use that metal for making jewellery / weapons because metal itself has its value / usages. (उस मुद्रा की धातु पिघला कर कुछ और चीजे भी बना सकते है. क्योकि धातु का अपना : अंतर्भूत मूल्य / उपयोगिता है.)	So currency notes have no intrinsic value. (उसके कागज का उपयोगिता कुछ नहीं है, क्योकि उसपे छपाई कर दी है.)

10.6.5 Fiat money: fiat means legally binding command (वैधानिक मुद्रा):



Fiat Money MUST fulfil two conditions SIMULTANEOUSLY: दो शर्तों का एकसाथ पालन :

1. It **MUST** BE in the form of Physical Currency Coin / Physical Currency Notes / VirtualCoin/ DigitalCoin/ CryptoCoin which can measure value **AND simultaneously और साथ ही साथ**
2. It **MUST** BE issued by the order of a King / Queen / Government / Central Bank.

	⟹ Govt issues all coins upto ₹ 1,000, using powers of powers of Coinage Act 2011. ⟹ Presently, govt issues Rs.50 paisa to Rs.20 coins. ⟹ Rs.20 coin: 12-sided Polygon shape (Dodecagon आकार: बारह बाजुओ वाला चतुष्कोण है) ⟹ ₹ 1 Note signed by Finance Secretary (वित्त सचिव) ⟹ ₹ 1 note doesn't contain "I promise to pay bearer…"
	⟹ **RBI issues:** Currency notes other than ₹ 1 Note under RBI Act 1934. ⟹ Currency notes have Governor's sign: *"I promise to pay bearer…"*

FAQ: why is it like this!!!??? WHY CAN'T RBI ISSUE BOTH coins & currency notes!!?

Ans. Send your suggestion to PM, to merge above two laws. Until it's implemented, remember table for passing exam. प्रधानमंत्री को कागज लिखो- दोनों कानूनों का एकीकरण करने के लिए. जब तक वह नहीं होता handout रटो।

10.6.6 Legal Tender (वैधानिक निविदा)

To become a 'legal tender', a given coin/currency MUST fulfil two conditions SIMULTANEOUSLY: एक साथ दो शर्तों को पूरा करना होगा.

1. It must be a FIAT MONEY (जो हमने ऊपर के सेक्शन में पढ़ा). AND SIMULTANEOUSLY और साथ ही साथ
2. It must be legally valid for all debts & transactions throughout the country. Other party can't refuse to accept. (लेनदार उसको अस्वीकार नहीं कर सकता)

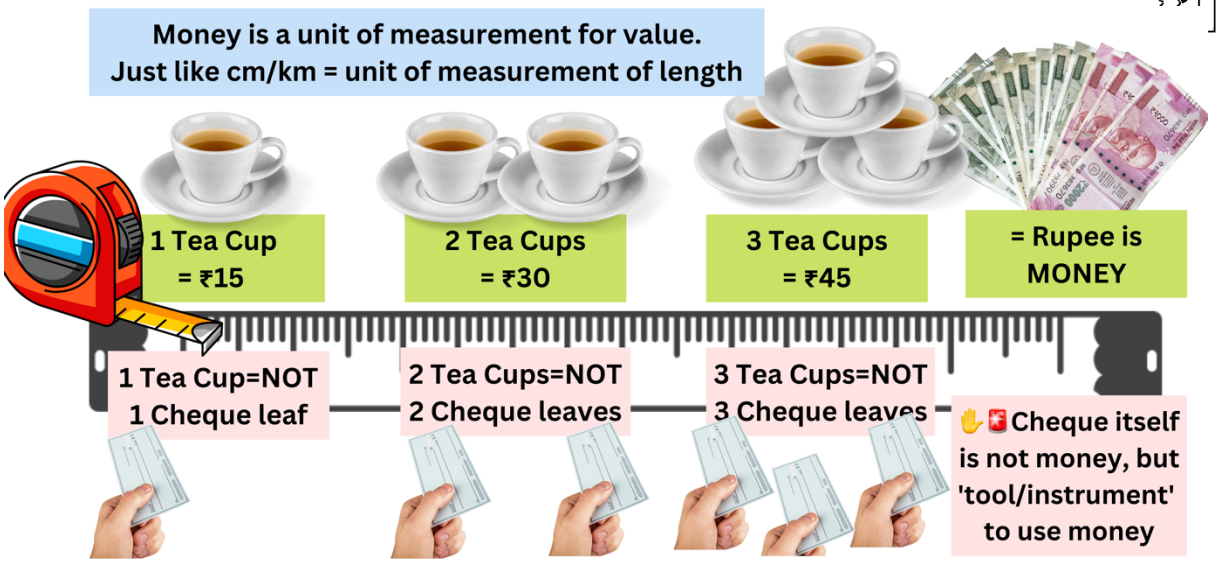


Table 2: fiat money and legal tender in India

ITEM	A) Is it in form of a physical coin OR physical Note OR Digital Coin that can measure 'value'?	B) Is it issued by Central Bank / Govt / King / Queen	C) Result: Is it Fiat money? (YES If Both Cell A+B give ✅+✅)	D) At present, is it legally valid for transaction in India as per Coinage Act or RBI Act?	E) Is it LEGAL Tender in India (Yes, IF C+D give ✅+✅)
₹ Indian Rupee: Coins & Notes, CBDC	✅	✅	✅	✅	✅
Demonetized (विमुद्रीकृत) ₹500,₹1000	✅	✅	✅	✋	✋
Dollar$, Yuan, Yen, Euro etc	✅	✅	✅	✋	✋
G-Sec, T-Bill (what is this-Ans. explained in video)	✋	✅	✋	✋	✋
Shares, Bonds, DD, Cheque, ATM, Cards, Kirana coin, Casino coin, Filmstar-Notes	✋	✋	✋	✋	✋
Marshall Island: SOV Venezuela: Petro coin	✅	✅	✅	✋	✋
Bitcoins	✅	✋	✋	✋##	✋

⇒ ##Bitcoin is not a Fiat money, yet it is legal tender in El-Salvador nation. But it is an exception. (बिट-कोइन का अल-साल्वाडोर देश में वेधनिक निविदा होना केवल एक अपवाद है, वो पूरी दूनिया के लिए नियम/व्याख्या नहीं है।)

⇒ Commemorative Coins = Fiat money yes but not legal tender unless notified by RBI/Government to be used as legal tenders. (स्मृतिचिन्ह रूप सिक्के- सामान्य भुगतान में इस्तेमाल नहि हो सकते, सिवाय के सरकार/आरबीआई उसकी अनुमति दे।)

😤 **FAQ: we can use cheque / credit card to pay fees. So, why are they not fiat money/legal tender?**

😤 **FAQ: I did not understand this thing and/or I want to engage in PhD/lawyer debate why "X" thing is legal tender?** Ans. watch lecture's recorded video again at 1X speed, more peacefully.

10.6.7 Legal Tender (वैधानिक निविदा)- Limited and Unlimited

Table 3: Fiat Money: Two types of legal tenders

A) Limited legal tenders (सीमित वैधानिक निविदा)	B) ∞ Unlimited legal tenders: (असीमित वैधानिक निविदा)
Recipient can refuse accepting payment beyond a limit (एक हद से ज्यादा छूटटे पेसे स्वीकारने से सामने वाला मना कर सकता है)	No such restriction.**
⇒ Coinage Act 2011 ⇒ Paper Note: Rs. 1 (एक रुपए की काग़ज़ी नोट) ⇒ Coins Rs. 1 / above ⟶ upto Rs.1000 ⇒ Coins 50 paisa x 20 coins ⟶ upto Rs.10 ⇒ Below 50 paisa coins are withdrawn (in 2011) So, 10 paisa, 25 paisa are ✋not legal tenders. (10 पैसा और 25 पैसा वैधानिक निविदा नहीं है। 50 पैसा वैधानिक निविदा है)	⇒ RBI Act 1934 - Section 26: Every bank note is legal tender in India. ⇒ They can be used for settling any amount of debt in India**, lender can't refuse to accept it.

Although in real life, Finance Act 2017: "Cash transactions for less than Rs.2 lakh only. Beyond that use Cheque, DD, NEFT etc. else penalty." This rule is to discourage tax-evasion / black money. For more Ref: Pillar#2. But for MCQs, simply accept that* ***bank notes = unlimited legal tenders. (हालांकि वास्तविक जीवन में काले धन / करचोरी रोकने वास्ते नगदी लेनदेन पर कुछ सीमा के बाद रोक है, किन्तु MCQ के लिए जो पढ़ाया वो स्वीकारो)

FAQ# my shopkeeper doesn't accept 50 paisa. Then how it is legal tender? मेरा दुकानदार नहीं लेता 50 पैसा का सिक्का। तो आप कैसे कह सकते है वो वैधानिक निविदा है?

Ans. as per RBI notification, 50 paisa is legal tender. So, accept it for MCQ & move to next topic. Further, feel free to file police case against shopkeeper, if you have plenty of free time/outrage for it.

FAQ ₹1 one rupee paper note is limited legal tender or unlimited legal tender? Ans. Read above table carefully. it's given in it in the left column.

MCQ: Which correctly describes the meaning of legal tender money? *(Pre-2018)*

(a) The money which is tendered in courts of law to defray the fee of legal cases
(b) The money which a creditor is under compulsion to accept in settlement of his claims
(c) The bank money in the form of cheques, drafts, bills of exchange, etc.
(d) The metallic money in circulation in a country

MCQ: Consider the following statements *(Asked in UPSC-CDS-2011-II)*

1. In India the minimum denomination coin acceptable for transaction is 50 paise.
2. Coins below 50 praise is not a legal tender for payment.

Which of the statements given above is/are correct?

(a) Only 1 (b) Only 2 (c) Both 1 and 2 (d) Neither 1 nor 2

10.6.8 RBI's App to help the blind identify currency notes (नेत्रहीन/ दिव्य-चक्षुओं के लिए)

80 lakh people in India are blind . To help them identify currency notes:

⇒ Old series of Gandhi notes have square (50), triangle (100) shaped markings.

⇒ New series of Gandhi notes have bleed lines, raised printing of Gandhi etc.

⇒ But, so many varieties of notes, difficult for VH persons to remember size & shapes so,

2020-Jan: RBI launched 'Mobile Aided Note Identifier (=MANI)' App.

⇒ Mobile camera scan & identifies the note (Even if it's half folded), gives audio notification in Hindi/English, also works offline

⇒ But can't validate whether note is genuine or fake (Counterfeit जाली नोट नहीं पकड़ सकता यह ऐप),

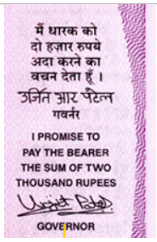
⇒ If a user is both blind and deaf, the app will tell him by giving vibrations.(मोबाइल कंपन)

10.6.9 Currency related General Knowledge:

Which note has what colour, which note has Red Fort / Sun Temple photo etc? General Knowledge Less important for UPSC-exams, more IMP for Non-UPSC Exams. So I'm removing it from Handout. You can learn such general knowledge from https://paisaboltahai.rbi.org.in/

New rupee symbol	₹: selected through a competition by Dept of Economic Affairs in 2010. Designed by **D.Udaya Kumar**, a Professor @IIT Guwahati. **Controversy-2025:** Tamil Nadu govt used Tamil letter 'ரூ' instead of '₹' symbol in its State Budget logo.
Nation Symbol	- Lion Emblem: Sarnath Pillar, Uttar Pradesh से लिया गया है - "Satyamev Jayate": slogan from Mundaka Upanishad से लिया गया है
Languages	While Constitution's 8th Schedule has 22 languages, but currency note has only 17. (15 on Reverse + Hindi & English on frontside (obverse) = 17) 🤯FAQ: Why it is like this? Ans. Either some Indian languages have similar fonts and/or send suggestion to RBI.
Museum	museum on numismatics and trade at Kolkata. (सिक्काशास्त्र और व्यापार का संग्रहालय)

10.6.10 "I promise to pay the bearer…": Meaning of this promise?



Colonia era: Promised to convert into full bodied gold or silver bars / coins worth the equal value in weight. E.g. 1 US Dollar = ~14 grams of gold. 1 British pound = ~73 grams of gold. ***(in my handout '~' means 'approximate')***

⇒ **Modern era:** Conversion into other Bank notes and "token coins" of equal face value. ₹ 2000 = 500 x 4 Nos.= 100 x 20 Nos etc.

⇒ Not inflation adjusted value. Not linked with weight of gold or silver.

⇒ Currency note is zero interest, anonymous bearer bond / Promissory Note.

10.6.11 Currency Note is not a promissory note?

Doubt: according to some website, currency note is not a promissory note under the negotiable instrument act. then what is write?

Doubt	as per academic books on economy?	as per the law (Negotiable Instruments Act)
Is currency note a promissory note?	Yes it has the features of promissory note.	No. It is not.

FAQ: Then what should I tick in MCQ? Ans. Depends on the context and options given in the MCQ.

10.7 DEMONETIZATION OF FIAT MONEY (विमुद्रीकरण: PREVIOUSLY 1946, 1978)

Definition? Demonetization means the REMOVAL of legal tender status of currency notes. (किसी मुद्रा की वैधानिक-निविदा के दर्जे/ क़ानूनन वैधता को रद्द करना- उसे विमुद्रीकरण या नोटबंदी कहते हैं)

⇒ 8/Nov/2016: Finance Ministry → Dept of Economic Affairs → notification to ban the "**Specified Bank Notes" (SBN)** of Mahatma Gandhi series ₹ 500 and 1000.

⇒ **Specified Bank Notes (Cessation of Liabilities) Act 2017**= It says: (1) RBI not required to honor "I promise to pay…" promise for 500-1000 ke banned notes. (2) It is illegal to keep banned notes (500,1000) except for research or numismatics-hobby or museum (with some terms-conditions).

⇒ SC Judgement on Demonetization ? Ans. REFER *pillar#2: taxation → black money*.

10.8 ₹2000 WITHDRAWN BUT CONTINUED AS LEGAL TENDER

2016-Nov: RBI introduced ₹2000 note, to meet the currency requirement of the economy in an faster manner after demonetisation of ₹500 and ₹1000 (नोटों की किल्लत पूरा करने के लिए इसको छापा जा रहा था)

- 2018–19: printing of ₹2000 banknotes was stopped.
- 2023-May-23rd: RBI ordered people to get it exchanged with other notes.

10.8.1 ₹2000 NOT DEMONETIZED but WITHDRAWN from circulation

- **2023-June: Reporter asked :** *Will the RBI withdraw the legal tender of ₹2000 note after September 30? (इसका विमुद्रीकरण नहीं हुआ. बस चलन में इसे हटा दिया गया है)*
- **RBI Governor replied**: *"There is no such idea."* That means

Question	interpretation
is ₹2000 fiat money?	Yes
is ₹2000 legal tender?	Yes
is ₹2000 in withdrawn from circulation?	Yes
is ₹2000 DEMONETIZED?	NOT YET.^^

- ^^As of 2025-Jan: this is the status, as per RBI official website

10.8.2 ₹ 2000 withdrawn- WHY? (क्यों चलन से हटाया गया)

We will only prepare the official reasons given by the RBI.

✓ 2000 Notes were issued as a temporary measure to address shortage of currency. Now there is adequate supply of other currency notes e.g. 100/200/500.

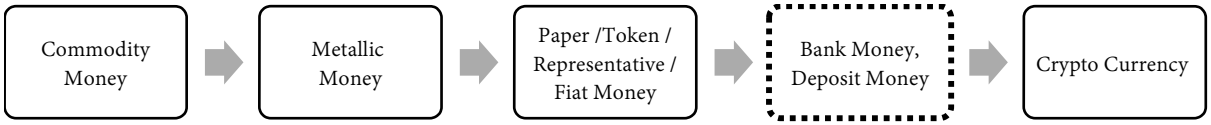
- ✓ This note is not commonly used for transactions. (वैसे भी भुगतान में इसका उपयोग कम ही होता है)
- ✓ RBIs clean note policy: to ensure availability of good quality banknotes to the members of public.
- ✓ 2000 note's paper has a lifespan of 4–5 yrs. Since most of these notes were printed in 2016–2017. So by 2022–23 its paper quality may start to deteriorate. (चार पांच वर्षों में कागज वैसे भी खराब हो जाता है)
- ✓ THERE IS NO MENTION of fighting black money/ corruption / fake notes by ISI-Pakistan.

10.8.3 💰🤮 Gresham law: good vs bad money | soft vs hard currency, Zimbabwe's Gold Coin

- ✓ Wait till we reach Pillar 3B: currency exchange. Presently, we're in Pillar 1A1.
- ✓ If unable to keep patience, watch https://youtu.be/x7rcf13JGh8?si=rkIwvohe38Gbw_FR

10.8.4 🤫💸👜⚪ DELETING: Redenomination: Iran's Toman Currency

Don't find it imp anymore so deleting.



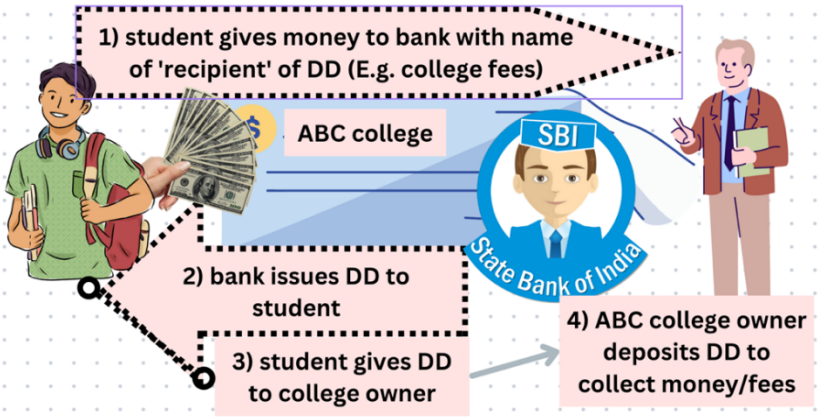
10.9 🐷🪽 BANK MONEY / DEPOSIT MONEY

10.9.1 🐷🪽 Paper orders: Cheque, Demand Draft (DD)

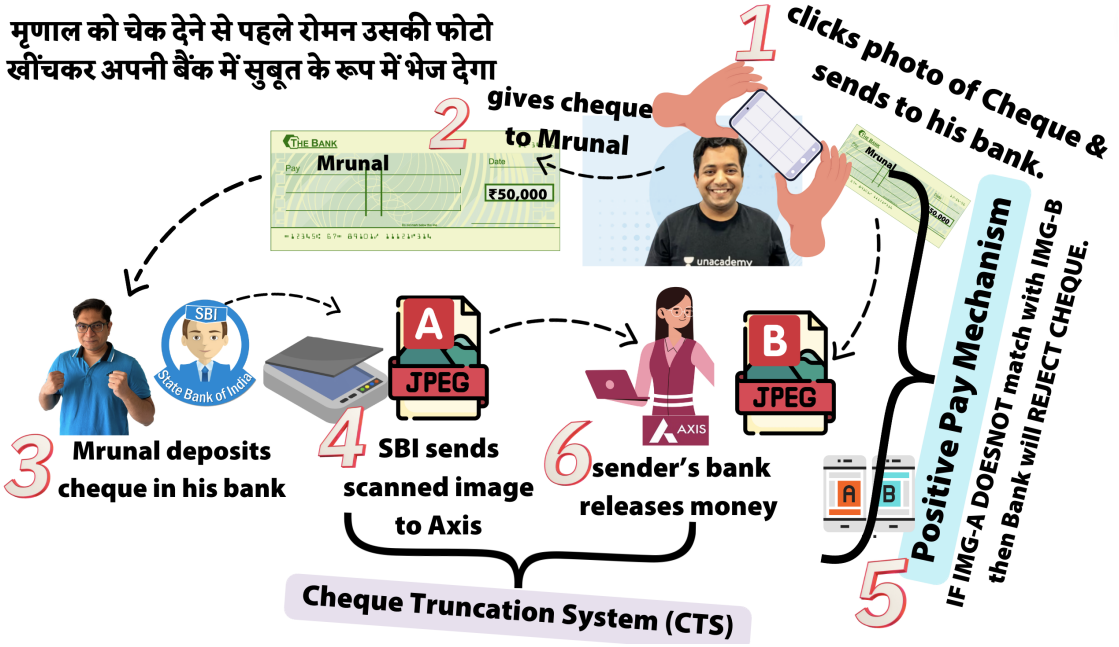
Law	Viceroy Ripon's Negotiable Instruments Act, 1881: Primary Objective? Cheque dishonor & forgery (परक्राम्य लिखत अधिनियम: बेंक खाते मे अप्राप्त रकम के चलते चेक अस्वीकार होना/जालसाजी के ख़िलाफ़ क़ानून).
Cheque	- **3 parties in a cheque**: 1) Drawer (Sender), 2) Drawee (Bank), 3) Payee (Recipient) - **Types of cheques:** 1) **Stale**: Not deposited in 3 months of issuance (3 महीनों में नहीं डाला तो बासी/अस्वीकार्य होगा), 2) **Post-dated:** Deposit after a specific date, - **Open / bearer cheque:** No 'crossing'. Anyone can encash. Not linked with name of receiver. - 3) Ante-Dated: notimp for IAS exam.
IFSC	**IFSC code:** Indian Financial System Code- 11 alphanumeric numbers to identify the bank branch, just like PINCODE identifies an area. (कौन सी बैंक ब्रांच है उसे चिन्हित करता है)
MICR	**MICR code:** Magnetic Ink Character Recognition. 9 digits code written in Iron Oxide ink for automated clearance. (चुम्बकीय स्याही से चेक के नम्बर को जाँचने के लिए।)

10.9.2 🪽 Demand Draft and Overdraft

- **Overdraft**: When person has insufficient bank balance, still he may withdraw money from his account (as a loan). Such facility is called **Overdraft**. Pradhan Mantri *Jan-Dhan* account has *Overdraft* upto Rs 10,000/- with certain conditions. *(More in Pillar1D: Financial Inclusion).*
  - o **Overdraft** is mainly for short-term operating expenses. *(अस्थाई रूप से लघु अवधि के लिए कर्ज)*
  - o **Loans** is mainly for longer term higher value expenses. *(दीर्घ अवधि के लिए मोटी रकम का कर्ज)*



10.9.3 Cheque ⟶ NPCi's Cheque Truncation System (CTS) & Positive Pay Mechanism



NPCi's Mechanism	Benefit (first SEE THE DIAGRAM)
Cheque Truncation System (CTS)	If SBI sent the copy of cheque in post/courier = more time wasted. CTS IS FASTER because SBI sending scanned image online.
Positive Pay Mechanism	If Mrunal (recipient) manipulated the cheque numbers e.g. using magic-marker/eraser to write ₹5,00,000 in place of ₹50,000 YET Roman (Sender) will be protected because both images will not match ⟶ then cheque will be rejected by his bank.

⇒ **FAQ:** I still DID NOT understand this. Ans. Watch The video lecture again at 1x speed.

⇒ **FAQ:** I've 500 counter arguments about how James Bond villain can misuse it? Ans. NOTIMP

10.9.4 CTS - Faster settlement in just few hours (2024)

BEFORE	After 2024 reform
CTS clearance/settlement took upto 1–2 working days. Bcoz Bankers may not send 1–1 cheque's scanned copy to other bank.	Settlement within few hours.

BEFORE	After 2024 reform
Banker would do it in bulk/batches.	
called **T+1** Settlement Cycle	called 'settlement on-realisation'

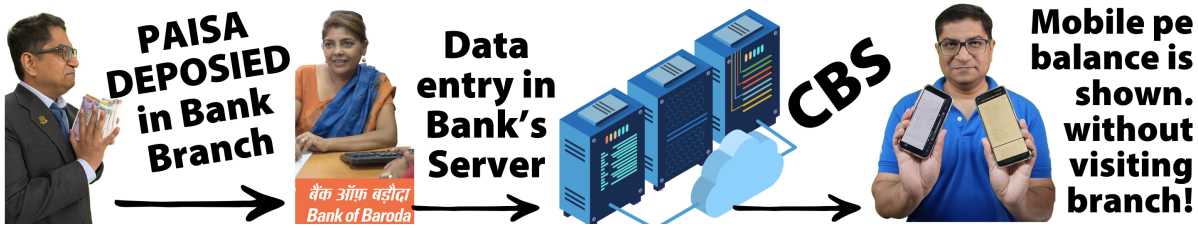
10.9.5 CTS ➡ 3 hr settlement postponed (2026-Jan)

- From 2026-Jan: RBI wanted bankers to clear the cheques **within 3 hrs.**
- Project was called "Phase-2 of *Continuous Clearing and Settlement on Realisation*"
- but bankers still facing operational challenges, so RBI postponed it.

10.10 Electronic Orders / Digital Payment

Payment & Settlement System Act 2007 empowers RBI to regulate card payment, e-payment related products and services. (भुगतान और निपटान प्रणाली अधिनियम 2007)

10.10.1 CBS (full form: Core Banking Solution /system)



⇒ It is a banking software with web-platform for centralized data management & branch-less banking. E.g. Finacle software; BanCS software; E-Kuber (used in RBI)

FAQ: What is the difference between Core Banking Solution (CBS) versus Core Banking System (CBS)? Ans. Fundamentally same thing. Dictionary word difference= notimp4exam.

MCQ: The term 'Core Banking Solutions' correct term? (UPSC Prelims-2016)

1. It is a networking of a bank's branches which enables customers to operate their accounts regardless of where they open their accounts.
2. It is an effort to increase RBI's control over commercial banks through computerization.
3. It is a detailed procedure by which a bank with huge non-performing assets is taken over by another bank.

Answer Codes: (a) 1 only (b) 2 and 3 only (c) 1 and 3 only (d) 1, 2 and 3

10.10.2 online money transfer systems: NEFT vs RTGS vs IMPS

Table 4: Digital Payment Systems- transaction limits may differ, depending on banks

	RBI's centralised payment systems (CPS)##		NPCi's IMPS
Type	RBI's RTGS: Real Time Gross Settlement	RBI's NEFT: National Electronic Funds Transfer	Immediate Payment Service
Single transaction- how much allowed	Min ₹2 lakh to ₹2000 crores (SBI). Each banker may have different limits. It is enough to know	Min ₹1 to Max ₹25 lakhs (SBI). Each banker may have different limits. It is enough to know NEFT	₹ 1 to maximum ₹5 lakhs

	RBI's centralised payment systems (CPS)##		NPCi's IMPS
	RTGS for big amt	for small amts.	
👨‍💼♂ **Target customers**	Bizmen wanting high value transactions instantly	Ordinary retail customers (आम जनता के छोटे भुगतानों के लिए है)	
🚚 **Settlement mechanism**	🤩Instant settlement (झटपट/ तुरंत निपटारा)	Settles net amount between banks at interval of 30 min.	🤩Instant settlement
💰 Fees	- **Inward** (recipient): 0 fee - **Outward** (sender): RBI allowed bankers to charge upto ₹50 fees ; but SBI etc bankers kept it for free.	- BEFORE: fees yes - After RBI Reforms: ₹0 fees.	fee + GST (but most bankers keep it free for small transactions)
⏰ **Timing**	BEFORE: only weekday office hours. After RBI reforms = available 24/7.		24/7 on all days (दिन रात अविरल रूप से)
👨✈ Who can provide this facility?	- **BEFORE-2021:** Only Banks can provide this facility - **2021:** RBI announces that non-bank entities can also become members of centralized payment systems (CPS = NEFT & RTGS). - So, in future, Amazonpay, PhonePe etc could also directly allow this		Banks + mobilepayment apps like Phonepe, Mobikwik etc.

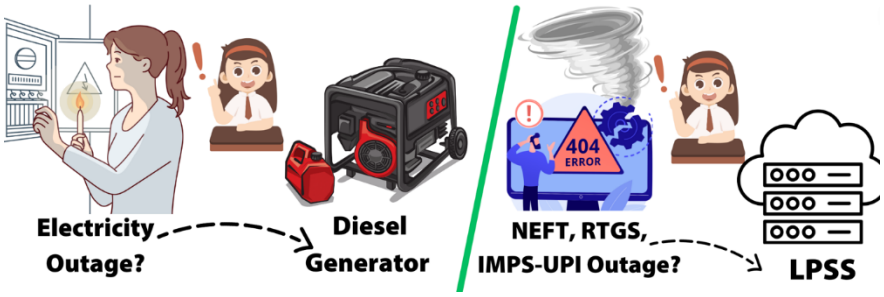
⇒ ##FAQ: 🚩👶🎓 **Why doesn't RBI Merge RTGS & NEFT?** Ans. RBI may have some technical operational reasons for it. if u feel morally-outraged then send suggestion to RBI Governor. Until implemented, memorize the facts as given in handout.

🔠 **MCQ. Find correct about RTGS and NEFT: (Pre'25)**

I. In RTGS, the settlement time is instantaneous while in case of NEFT, it takes some time.
II. In RTGS, the customer is charged for inward transactions while that is not the case for NEFT.
III. Operating hours for RTGS are restricted on certain days while this is not true for NEFT.
Codes: (a) I only (b) I and II (c) I and III (d) III only

Note: RBI had reformed fees on NEFT/RTGS from **2019**. And since then I have been teaching it every year. UPSC finally asked in **2025** Prelims 6-7 years after the event. So **yes, handout is lengthy,** because otherwise you will not be able to solve such MCQs.

10.10.3 📡 LPSS- lightweight payment and settlement system



- RBI's (emergency/backup) alternative to NEFT, RTGS and IMPS/UPI.
- To provide a temporary & portable solution during natural disasters and war. (प्राकृतिक आपदा या युद्ध के चलते बाकी प्रणालियाँ बंद हो जाए तो अस्थायी रूप से इसको इस्तेमाल कर सकते)
- Can be operated from anywhere by a bare minimum staff. How it works NOTIMP.

10.10.4 🕵️ RBI: beneficiary account name look-up facility for NEFT / RTGS (2024)

Can u look up name of the beneficiary b4 sending ₹₹	Before	After 2024-Oct Reform
UPI/IMPS	Yes	Yes
NEFT & RTGS	No	Yes**

- **When the remitter (sender) enters the beneficiary's bank account number and IFSC code → beneficiary's name will be displayed to the sender. (नंबर टाइप करके नाम चेक कर सकते हो।)
- 🤩 Benefit?: minimizes errors & frauds in money transfers.

10.10.5 💸📡 DELETING RBI's Next Generation Treasury Application (NGTA), 2020-Oct

Deleting from handout. Because yet to be implemented. This topic didn't develop much after 2020.

👨‍🏫 **Mrunal comments before next section**: In the digital wallet apps → companies introduce new features & RBI introduces new technical norms on daily basis. We are not here to write a MBA research paper on their business-model & RBI regulations. Q.*"sir u said x thing can't be done in Paytm App but now it can be done…as per my experience!! Ans. I've given simple illustration to explain thing. They keep adding new features in the App.* UPSC asking only general broad MCQs from this. #🕰️थोड़ा-पढ़ो-आगे-बढ़ो

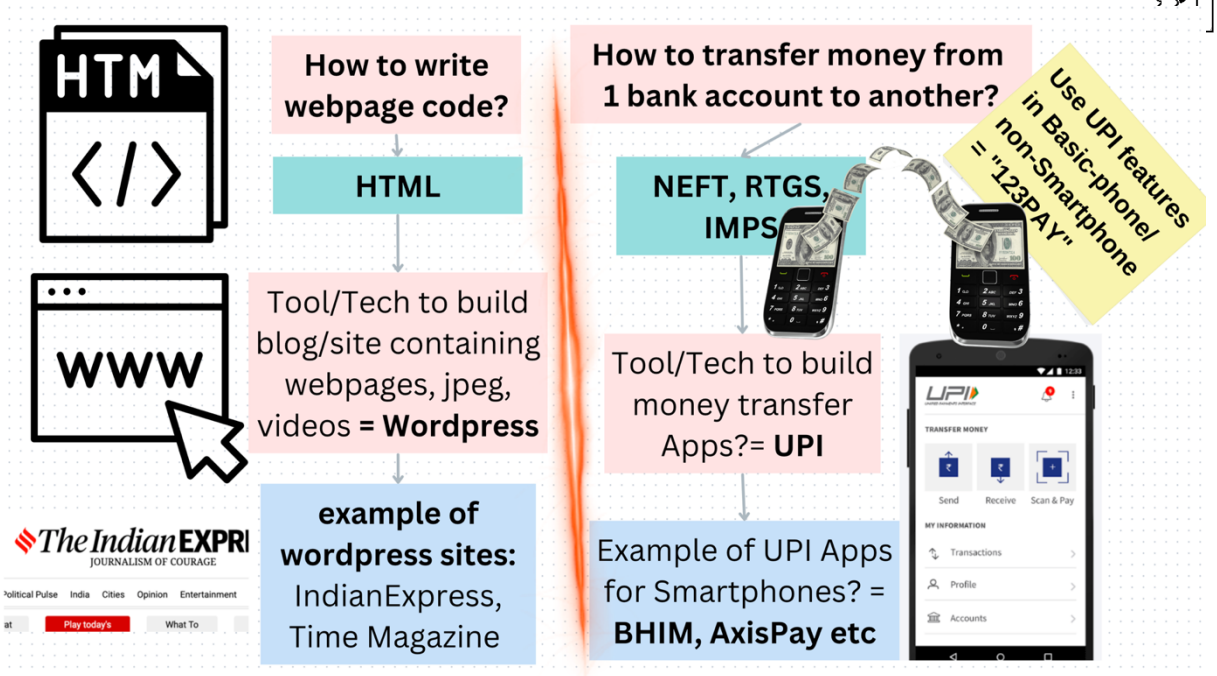
10.11 🕵️ National Payment Corporation of India(NPCi)

⇒ 2008: NPCi is registered under Company Act as a "Not for Profit Company". (मुनाफा रहित कंपनी)
⇒ Originally it was founded by 10 banks with ₹ 100cr capital. Later on in 2020: Paytm, PhonePe, Amazon Pay etc also became shareholders of NPCi (ये सब शेयर धारकों ने पूंजी डालकर इस कंपनी को बनाया है)
⇒ Objective? cost-effective payment solutions / technology for Banks. (रियायती लागत पर बैंकिंग टेक्नोलॉजी)
⇒ Its UPI-BHIM app & other mechanisms also help in financial inclusion of villagers & poors by expanding the reach of banking network. (गरीबों / गाँववालों के वित्तीय समोवेशन में मदद के लिए अलग अलग तकनीकी आविष्कार किए है) ✅ More abt financial inclusion in 📄Pillar# 1D.

🔠 **MCQ-EPFO-2023: Consider the following statements:**

- Statement I: In India, the RTGS and NEFT payment systems are owned and operated by National Payment Corporation of India (NPCI).
- Statement II: National Payment Corporation of India is an entity promoted (=owned) by banks.

(a) Both right, st2 explains st1 *(b) Both right, but st2 doesn't explain st1*
(c) St1 is correct but St2 is wrong. *(d) st1 is wrong but st2 is right.*



Above is not 100% technically correct explanation, but it is just to help you visualise the concept.

10.12 🕵🏻 NPCI → UPI (Unified Payment Interface)

It is a technology for building digital payment apps based on IMPS with following features:

⇒ QR Scan & Pay to merchants.

⇒ You can link bank account for direct transfer of money without storing money in 'wallet' first. (unlike AmazonPay / Mobikwick etc)

⇒ Such app can have Push transaction (e.g. Remittance to family), Pull Transaction (e.g. monthly bill deduction by Electricity Co.) or even Bill sharing among friends.

⇒ **Examples of UPI based app:** SBI's SBIBuddy/Yono, Axis Bank's AxisPay and NPCi's BHIM.

⇒ **UPI 2.0 (2018):** Upgraded version of UPI. What are its new features? How is it different than UPI 1.0? Ans. NOTIMP #RAFTAAR 🏎️🐢

⇒ NPCI has been signing agreements with organisations in Bhutan, Nepal, Singapore, etc. so that directly Indian tourists in those foreign countries can directly spend the money from their Indian bank account by using UPI-Based Apps.

10.12.1 💪 UPI: Push vs Pull Transactions

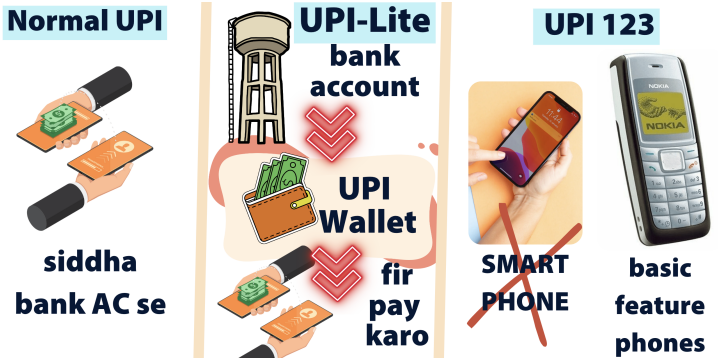
UPI based apps allow for both 👉Push transactions and 👉Push transactions.

👉Push transactions	💪Pull transaction
👉Push = when u send money by your own discretion e.g. sending money to a relative/friend based on your mood. (आपकी मर्ज़ी या विवेक से आप ख़ुद जो पैसा भेजते है।)	💪Pull transaction = when you have given command that every month electricity company should cut the money from my bank account → company will pull money without requiring u to push/click buttons every month.

🚩👶🏻FAQ: Beyond that, whether "X" thing is push/pull?= Ans. NOTIMP #RAFTAAR 🏎️🐢

10.12.2 UPI Delegated Payments to create secondary accounts (2024-Aug)

e.g. Father (primary account-UPI) creates a secondary account-UPI daughter. Then she can use money from father's bank account upto Rs.X limit using her mobile.



10.12.3 Transaction limits- Normal-UPI:

Max amount	Category of daily transaction
₹1 lakh/day	Normal transactions.
₹ 2 lakhs/day	Capital Markets (e.g. Mutual Funds) Credit card payments, Loan re-payments/ EMI, Insurance premium/fees payment etc.
₹5 lakh/day	⇒ Purchase of G-Sec under Retail Direct Scheme, IPO Subscriptions. What is the meaning of the term? (Ref: Pillar#1C) . Medical and educational fees. ⇒ Tax Payment to Govt (2024-Aug onwards)

10.12.4 Transaction limits- UPI123, UPI Lite: NOT IMP

Not important. RBI frequently changes them so poor cost-benefit memorizing it.

10.12.5 UPI-Lite (on-device wallet)

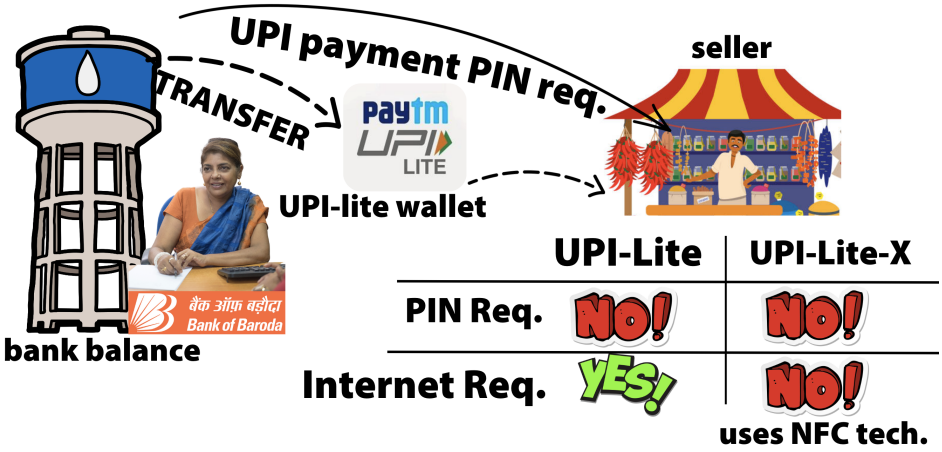
User adds money from his bank account to his UPI Lite wallet on his mobile App. (e.g. Google Pay App, BHIM App)

Benefits of UPI-Lite wallet?

- Faster. Because No need to provide UPI-PIN number during transaction.
- Less chances of transaction failure- even if sender's bank server is facing technical issues.

10.12.6 UPI Lite vs UPI Lite-X (offline payment)

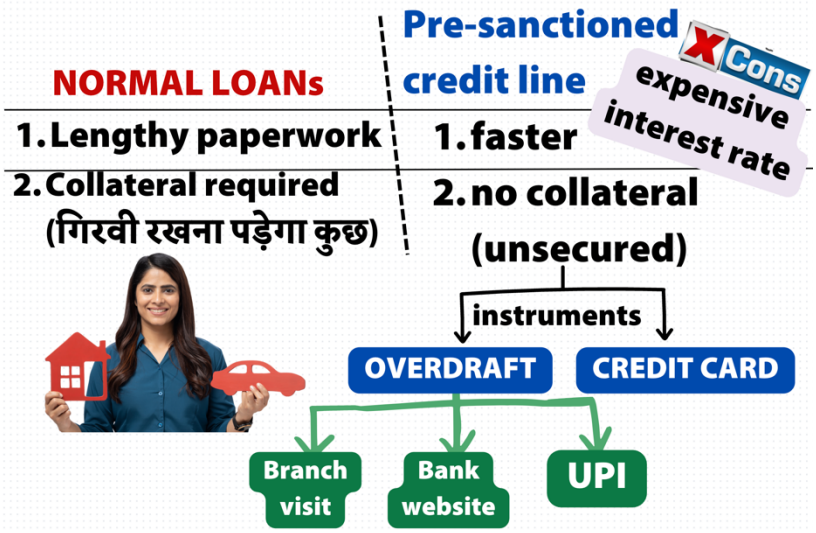
UPI Lite	UPI Lite X
Even if sender's bank server down it works / if Payment App unable to connect with Core Banking Solution (CBS) of bank immediately.	same.
However the App needs internet connectivity.	doesn't need internet. Works offline using **Near Field Communication (NFC)** technology. But, User has to come online after some days for data synchronisation.



10.12.7 Conversational payments in UPI

- It will help users to chat/converse with AI-powered systems to make payments. e.g. similar to how Smart speakers function: *'Hey Siri/Alexa/Google remind me to buy milk on X date.'*
- **Benefit?** Voice command make it more convenient for **blind** persons to use UPI.

10.13 Pre-sanctioned credit line



When a banker allows an account holder to borrow upto a predetermined amount of money, without having to apply for a loan, it is called Pre-Sanctioned Credit Line. (इसका हिंदी करने का मतलब नहीं है. मुख्य परीक्षा में इस प्रकार की चीजें नहीं पूछते)

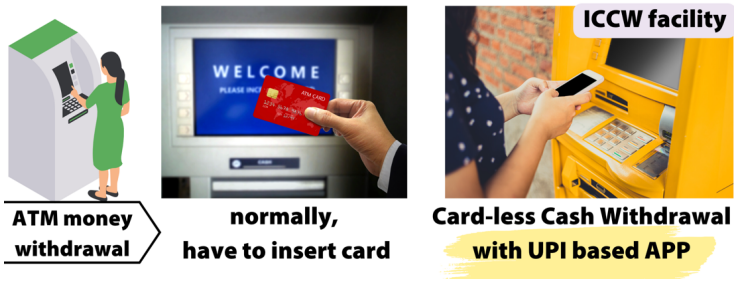
10.13.1 UPI: Pre-sanctioned credit line/Overdraft in UPI (2023-Apr)

- RBI allowed Pre-Sanctioned Credit Line / overdraft facility extended to UPI.
- So, even if a person does not have balance in his bank account, still he can PAY using UPI while shopping. (and then later, when he gets the salary/pension/income deposited in his bank account, then the banker will recover the principal and interest.

10.13.2 🔪👜 UPI: Benefits of UPI-overdraft facility= BNPL / buy now pay later

- Help customers in "Buy now pay later (BNPL)" format of shopping. (Ref: Pillar#1B1 for more)
- Poor people require small small amount of money/loans during last weeks of a month. They cannot travel to the city's bank branch for the paperwork and banker may take many days to pass application. Consequently, poor villagers borrow from the local moneylender/Shroff/Baniya at high interest rate. UPI-Overdraft can help overcoming this to some extend. = helps in Financial inclusion of the poor. (गरीबों के वित्तीय समावेशन में मदद)

10.13.3 👜 UPI: Interoperable Card-less Cash Withdrawal (ICCW)



- Benefit? physical card not inserted in ATM machine → ↓ dangers of card cloning, etc scams.

10.13.4 🪙🏧 UPI: facility for depositing /withdrawing money

- Scan QR code➡ transfer money from bank account via UPI ➡to get coins from machine
- Deposit cash in Cash Deposit Machines (CDMs) by logging into your bank account via UPI (no need to insert debit card in the machine)
- Withdraw cash from ATM machine by logging into your bank account via UPI (no need to insert debit card in the machine)

10.13.5 📲 UPI new safety rules from 2026 June

Matter	Before	After 2026-Jun reform
High value transaction extra layer of safety	PIN number alone is enough to send money	PIN + biometric(FaceID/Fingerprint) etc. required for extra safety
Pre-Transaction Name Verification	sometimes recipient's real name is not shown.	UPI apps must mandatorily display the verified name of the recipient before the sender sends money- to avoid scams.

10.13.6 NPCI → UPI → BHIM Full form: Bharat Interface for Money (2016)

⇒ Money transfer app designed by NPCi. It works on Android, iOS/Apple mobiles as an APP,

⇒ Works even on non-smart (=basic feature) phones using *99# USSD- Unstructured Supplementary Service Data.

⇒ Bank to bank / peer to peer transaction using mobile phone (xyz@upi). No need to install multiple apps for each bank account (SBIBuddy, AxisPay etc) just one BHIM app to use all such bank accounts.

⇒ (1) App has **3 factor** authentication system- as per Indianexpress **(2) App has 2 factor authentication system** as per NPCi official website. (FAQ: then which fact is true? Ans. i'd go for official website.)

⇒ Your money stays in bank account and earns interest. It's not stored in 'wallet outside your bank account' as it happens in Mobikwick, Phonepe etc. (PS: in later stage, PhonePe etc private apps also included UPI type mechanism so storing money in 'wallet' became optional but we need not engage in Hairsplittery/PHD.)

⇒ No cards involved so no MDR or such hidden charges.

MCQ. Find Correct Statement(s) [Prelims-2018]

1. BHIM app allows the user to transfer money to anyone with a UPI-enabled bank account.
2. While a chip-pin debit card has four factors of authentication, BHIM app has only two factors of authentication.

Answer codes: (a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2

10.13.7 BHIM 3.0 (2025) with new features

- Now available in more than 15 Indian languages for better accessibility.
- Optimized to work in slow internet; Add family / friends in the group to split bills.
- Tools for tracking expenses/budget/ monthly financial planning.
- Built-in reminder tool for monthly bills; UPI-lite wallet gets better user experience.

10.13.8 BHIM VEGA merchant tool, pay without switching to 3rd party App

BEFORE	AFTER
You placed a food order on the Swiggy app → Go to the third party app e.g. PayTM app for payment.	If merchant enabled BHIM Vega facility then customer can directly pay from the merchant's app without having to switch to a 3rd party app.

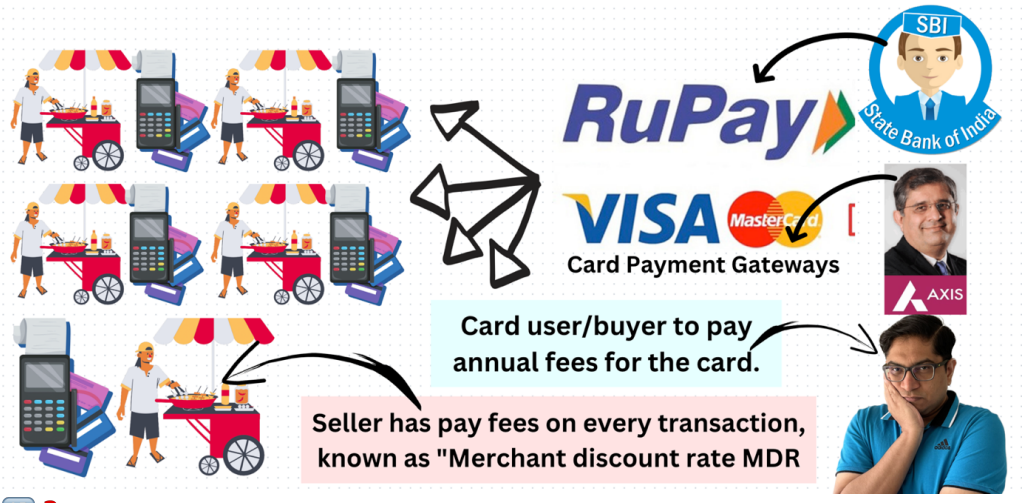
10.13.9 NPCI → Rupay Card Payment Gateway (2012)

⇒ Rupee + Payment = RuPay card is world's 7th payment gateway similar to Mastercard, Visacard, China's Union Pay. Rupay charges less fees than other companies.

⇒ Works in **3 channels:** 1) ATM, 2) Point of Sale Device (PoS/card reader machine), 3) Online portals.

⇒ Rupay debit card given free with PMJDY (Jan Dhan) Bank account (Refer Pill#1D).

⇒ Rupay has signed collaboration agreements with some digital payments companies in Singapore, Japan etc. So, it may be used in those countries. But presently Rupay is not having the widest reach across all the countries unlike MasterCard and Visa.



MCQ. Find Correct Statement(s) [Prelims-2017]

1. NPCI helps in promoting the financial inclusion in the country.
2. NPCI has launched RuPay, a card payment scheme.

Answer codes: (a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2

10.13.10 NPCI other notable initiatives

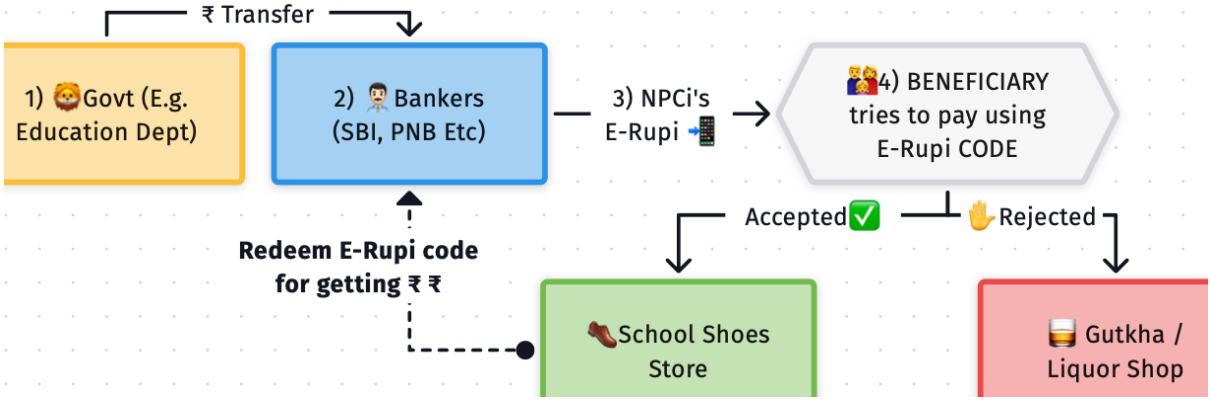
Table 5: NPCi has built following mechanisms / platforms / systems / apps:

BBPS & UPMS	- Bharat Bill Payment system for monthly payments of utility bills (gas, electricity etc.) - **2022-Jan**: NPCi 's daughter/subsidiary company Bharat BillPay Ltd. Company ⟶ launched a Unified Presentment Management System (UPMS) ⟶ but technical stuff. NOTIMP4 exam **FAQ: How is BBPS vs UPMS similar or different? Ans. NOT IMP 4 Exam. These are one-liner GK. Not PHD thesis topics!**
NFS	National Financial switch. This technology runs the ATM network
CTS	Cheque Truncation System learned in previous section.
NETC	⟹ National Electronic Toll Collection provides technology support to the FASTag toll collection. (राजमार्ग पर पथकर वसूलने के लिए)
PAI chatbot	Artificial Intelligence (AI) based chatbot to reply to people's online queries about NPCI's products like FASTag, RuPay, UPI, AePS etc.
DigiSaathi Helpline Number (14431)	RBI launched with help of NPCi. - for digital payment & card payments problems. The No. is 1800 891 3333 (not imp4exam but personal use)
DakPay (2020)	Payment App by Department of Post and its India Post Payments Bank
BharatQR(2016)	Given in next section of this handout. Scroll down
E-Rupi (2021)	Given in next section of this handout. Scroll down

10.14 Direct Benefit Transfer (DBT) ⟶ E-Rupi by NPCi (2021)

- (Origin) E-rupi is an SMS /QR Code based Prepaid Cashless Electronic Voucher created by National Payment Corporation of India (NPCI) using its UPI Platform.

- It is also supported by Finance ministry's Department of Financial Services and Health ministry's National Health Authority (NHA)

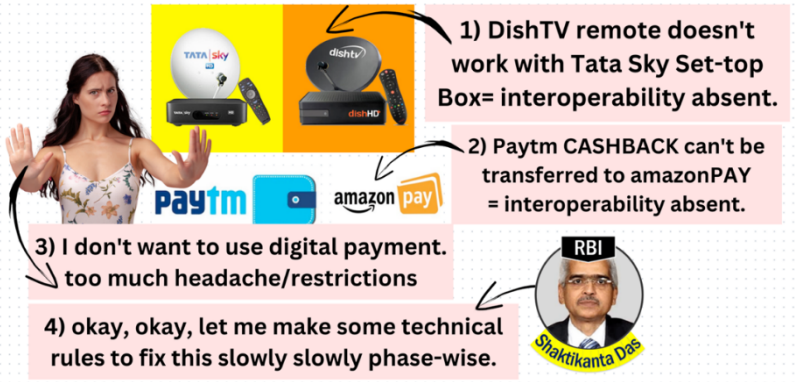


10.14.1 e-RUPI vouchers: scope expanded (2023)

	Before	After 2023
Who can Buy E-Rupi codes/vouchers?	1) Govt & 2) Pvt Companies	1) Government, 2) Pvt Companies AND 2) now Individual persons can also buy it (e.g. for gifting to their maid/driver etc.)
Who can SELL E-Rupi codes/vouchers?	only Banks can sell	1) Banks AND 2) Non-bank Prepaid Payment Instrument (PPI) e.g. AmazonPay, Mobikwick etc.
Max e-rupi code-value	₹10,000	₹1 lakh

FAQ: I still didn't understand this! Watch this Youtube Short-
https://youtube.com/shorts/1qoqwvDgbLA?si=VO_aJiHdzisJVCVL

10.14.2 Interoperability Problem (इंटरऑपरेबिलिटी)



FAQ: What are those technical Rules to improve interoperability? NOTIMP #RAFTAAR.

10.15 QR Codes: compulsory to implement UPI/Bharat QR from 2022

- 1990s: 2 dimensional (2D) Quick Response (QR) Code system developed in Japan.
- Indian E-payment systems function on 3 types of QR Codes, namely,

Table 6: Difference between UPI QR vs Bharat QR? NOT.IMP.4.EXAM थोड़ा-पढ़ो-आगे-बढ़ो

Proprietary QR	⟹ e.g. Paytm or PhonePe ke exclusive QR codes.

code	⇒ 😨Problem? Other companies' app may not be able to scan and pay it. ⇒ So, RBI has prohibited companies from launching any new proprietary QR code.
UPI QR code	Generated by UPI app of NPCi. 🤩It can be scanned by all Apps which are developed using UPI technology.
Bharat QR code	Developed by NPCi with help of Visa and Mastercard. 🤩It can be scanned by all Apps which are developed using UPI technology.]

⇒ RBI has ordered Paytm, PhonePe, MobiKwik, Google Pay etcto shift to UPI QR or Bharat QR

⇒ 🤩Benefit? All Payment apps will be able to scan these QR codes → 'interoperability'.

10.15.1 💸rMQR (Rectangular Micro QR Code)

⇒ 1990s: JAPANESE ENGINEER Masahiro Hara, designed the square QR code

⇒ Now he's working on rMQR (rectangular QR code). rMQR can be printed in narrow spaces

10.16 CREDIT & DEBIT CARDS (ALSO CALLED 'PLASTIC MONEY')

Table 7: Card Types based on Payment modality

1) Credit Card (कर्ज पे शॉपिंग)	2) Debit Card (अपने बैंक बचत का पैसा इस्तेमाल करो)
If Credit card used for withdrawing money from ATM, then it's a type of 'borrowing', so, bank levies interest rate.	Debit card can be used to withdraw ₹ ₹ from ATM from your existing bank balance. So, it's not 'borrowing'.
Credit card vs Overdraft? 1. Credit cards = for shopping. You carry card in your pocket. Bank may give u some reward-points/gifts if you spend "₹.X" per year 2. - for overdraft= you have to manually visit the bank branch, fillup form. It's not in the form of a 'card'. Doesn't give u points/gifts. ✋Further difference not.imp.not.here.for.IBPS-exam	**Debit card vs Overdraft?** You should be intelligent enough to self-understand that much. शिक्षक हर चीज की स्पून फीडिंग करेगा/चम्मच से खिलाएं वह विद्यार्थी के मानसिक विकास के लिए अच्छी बात नहीं

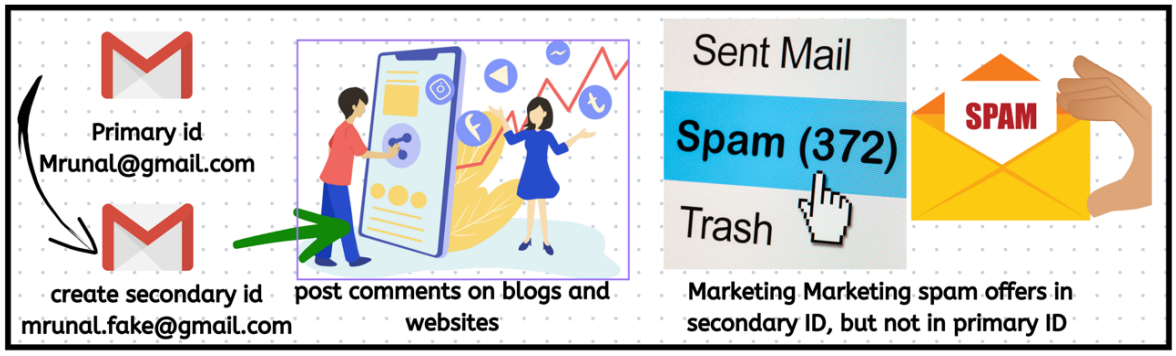
⇒ **3) Hybrid Card / Duo Card:** Single card with two chips for 1) credit card and 2) debit card. So, you don't have to carry two separate cards. E.g. Indusbank Hybrid card. एक ही कार्ड में दोनों का मजा🤩

⇒ **4) Pre-paid card:** It's a subtype of debit card. Debit card is linked to a given bank account. But Pre-paid card can be bought even without having account in the given bank e.g. **IRCTC's UBI Prepaid Card** can be used for buying rail tickets, meals etc. support provided by Union Bank of India (UBI) and NPCi's Rupay gateway. (इस कार्ड को पैसा देके खरीदना पड़ता है। बैंक खाता होना जरूरी नहीं)

10.16.1 Card Types based on Security Features: (सुरक्षा प्रणाली के हिसाब से कार्ड के प्रकार)

HomeWork: Read about RFID & NFC from Sci-Tech book/internet.

10.16.2 Card Tokenization (from 1/Oct/2022)



⇒ Tokenisation = creates a set of numbers called 'token' to replace/hide the sensitive card data such as the 16-digit account number, expiry date and security code. (हैकिंग से बचने वास्ते असली कार्ड नंबर की जगह टोकन/काल्पनिक-नंबर बनाकर दे दो एसी बात है। ताकि विक्रेता का आपके कार्ड की गुप्त जानकारी मिले नहीं)

⇒ This ensures your card information is not disclosed to merchant website= protection from hacking. Card owner can generate Tokenization through visiting card-issuer's/banker's website/App through laptops, desktops, tablet, wearables (smartwatch) etc.

⇒ From 1/10/2022, this became effective.

Figure 1: थप्पड़ से डर नहीं लगता, Identity theft se lagtaa hai!

10.16.3 Beyond OTP - additional factor of authentication (2026)

This is better understood through a case study

- if I purchase the Adobe Photoshop subscription from its USA website (= cross-border transaction) ➡ I type my credit Card number and submit SMS-OTP (=2 factors done) ➡ still, the banker may phone-call me for voice verification if bankers feel the transaction is suspicious. (=this is third layer or additional of authentication.)
- This additional authentication required based on (suspicious) transaction's behavior, location, device, history.

Matter	Before	From 1/April/2026
additional authentication required for suspicious transactions ➡	for international transactions.	for both international and domestic transactions (if they're suspicious)

This additional authentication made be done by

- Something you know: PINs, passwords, passphrases.
- Something you have: Registered mobile device, software or hardware tokens.
- Something you are: Biometrics (fingerprint, face ID, voice pattern etc.).

10.16.4 Credit Card Network portability order by RBI (2024)

Summarized in the graphics/meme. If not understood, watch lecture video.

10.16.5 CVC Less transactions

⇒ Normally, during online-transaction, customer has to enter Cardholder Verification Code (**CVC**) EVERY-TIME when making payment.

⇒ but after the RBI's new norms, this may not be required every time.

10.17 CARD/TECH FOR TRANSPORT / TRAVEL

10.17.1 National Common Mobility Card (NCMC, 2019)

- 'One Nation One Card Model' by **Ministry of Housing & Urban Affairs (MoHUA)** with the help of NPCi, Bharat Electronics Limited (BEL Company), and some banks. It helps developing special cards to pay tickets → Train station/train gate can open automatically, upon tapping the card, like in hollywood movies.

It has 3 components: (तीन घटक है इसके)

- 1) NCMC Card: To be issued by banks. Nature can be Debit/Credit/Prepaid. Can be used for metro, bus, suburban railways, toll, parking, retail shopping etc.
- 2) SWEEKAR: Automatic Fare Collection System. (यात्री किराया वसूलने की स्वचालित प्रणाली)
- 3) SWAGAT: Swachalit / Automated Gate at railway/bus station.

10.17.2 SBI's Nation First Transit Card (2023)

Enough to know it can be used for paying toll fees, parking fees and doing shopping.

10.17.3 FASTag/NETC for paying toll fees at highways (2017)

	FASTags are prepaid rechargeable tags that allow for automatic toll collection at electronic toll collection gates Radio-frequency Identification (RFID) technology.

⇒ 2017: National Highways Authority of India (NHAI, a statutory body under Highway ministry) launched 2 mobile apps **MyFASTag** and **FASTag Partner** to facilitate Electronic Toll Collection (इलेक्ट्रॉनिक माध्यम से पथकर/टोल वसूली के लिए राष्ट्रीय राजमार्ग प्राधिकरण द्वारा अलग अलग एप बनवाए गए हैं).

⇒ NPCi's has developed **National Electronic Toll Collection (NETC)** technology to aid this.

⇒ Benefit? faster traffic movement, tracking stolen cars.

⇒ From 2021, it became mandatory for all vehicles passing through tolls to have FASTags.

⇒ 2024-25: govt wanted to Fastag with satellite based toll collection but later postponed the plan. (Ref: Pillar#5 infra)

⇒ 2025: Annual pass available at ₹ 3,000. It will allow commuters to travel toll-free for 200 trips or one year, whichever comes first. (not imp for 4 UPSC but for non-IAS exams)

10.18 MDR: FULL FORM: MERCHANT DISCOUNT RATE (व्यापारिक छूट दर)

⇒ MDR is the fees that a merchant must pay to his (acquirer) bank for every card transaction.

⇒ MDR fees is **shared** among 3 parties 1) customer's card issuing bank 2) merchant's acquiring bank 3) payment gateway provider. (तीन पक्षों में इस फीस को बांटा जाता है)

⇒ Note – in above image, the %/amount is only for example/illustration. Real % are different.

⇒ MDR hurts merchants' profit margin, discourages them from adopting Point of Sale (PoS) terminals (=card swiping machine) ⟶ obstacle to digital economy. (व्यापारी के मुनाफे पर असर होता है इसलिए वह कार्ड मशीन अपनाने में उदासीनता दिखाते है।)

Table 8: Steps taken to address MDR issue

RBI	⇒ 2017: Imposed ceilings on MDR fees i.e. MDR can't be more than "x%" of the transaction amount. (रिज़र्व बैंक द्वारा तय सीमा से ज्यादा फीस नहीं मांग सकते)

10.18.1 💳 MDR Subsidy by MEITY

Instrument	UPI	NPCI's Rupay Debit Card	NPCI's Rupay Credit Card Mastercard/Visa's Debit/Debit card etc
MDR % on transaction value	0%	0%	"x"% (NOT ZERO)

⇒ 😢 So, if a banker promotes Rupay debit card / UPI App ⟶ Banker will not earn MDR fees. So bankers promote MasterCard & Visacard to their customers and merchants (instead of Rupay/UPI). (बैंकर को मज़ा नहीं आता क्योंकि उसे डेबिट कार्ड व यूपीआई एप में MDR की फ़ीस नहि मिलेगी।)

10.18.2 💁‍♀️FAQ: Does Rupay cards have MDR charge or not?

Read the terms given in handout carefully.

Rupay credit cards= Yes MDR | meanwhile, Rupay DEBIT cards= ZERO MDR.

🚩👶🎓 **FAQ: "I've 500 jaat ke PHD doubts & counter arguments about this MDR subsidy"** Ans.

Sure. Plz waste your precious time reading egazette.nic.in/WriteReadData/2021/231960.pdf

10.18.3 📲💰🦁 UPI Subsidy/ incentive by Govt for upto ₹2000 transaction value

- Boss? : 2021 onwards- The Ministry of Electronics and Information Technology (MeiTY)
- Boss? 2025 onwards: FinMin ⟶ Department of Financial Services

Merchant	UPI transaction value	MDR	Bonus by Govt to merchant
Small Merchant	Up to ₹2,000	0%	0.15% of value to merchant
Small Merchant	Above ₹2,000	0%	None
Large Merchant	Any Amount	0%	None

10.18.4 🦁 No Fees on Users for UPI Transactions – says Finance Ministry (2025)

- Newspapers run daily rumors on how there will be some fees on UPI transactions. But,

- 2025-Aug: Finance Ministry confirms no proposal to impose fees on UPI transactions.
- So whatever **clickbait videos /headlines** you see on internet. IGNORE THEM. This topic is over from exam point of view.

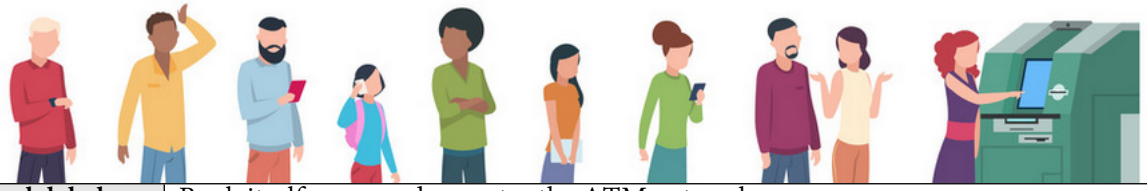
MCQ. What is "Merchant Discount Rate" ? (IAS-Prelims 2018)

A. The incentive given by a bank to a merchant for accepting payments through debit cards pertaining to that bank.
B. The amount paid back by banks to their customers when they use debit cards for financial transactions for purchasing goods or services.
C. The charge to a merchant by a bank for accepting payments from his customers through the bank's cards.
D. The incentive given by the Government, to merchants for promoting digital payments by their customers through Point of Sale (PoS) machines and debit cards.

10.18.5 ATM and its Types

⇒ Automated Teller Machine (ATM) allow bank customer to withdraw money or check balance without visiting any branch of a bank. (स्वचालित रूप से नकदी मुद्रा देने वाला यांत्रिक डिब्बा)

⇒ ATM network works on NPCi's NFS (National Financial Switch) technology.



Bank label	Bank itself owns and operates the ATM network.
Brown label	Bank owns but operations outsourced (cash refilling truck, guard) ⇒ ***Note:*** *Some books offer different definition, but I've taken above definition from Economic Times Article 2015.(कुछ किताबों में व्याख्या अलग है मैंने इधर से ली है)*
White label	- Non-Bank owns & operates. E.g. Muthoot Finance, Tata Comm, Prism - FAQ: What is on-tap licensing system for White-label ATM? Ans. Outdated topic for 2024-25.
Micro-ATM	Bankmitra manually makes entries of deposit and withdrawal for customer. He uses handheld device for using (debit or ATM) card & Aadhar biometrics (fingerprint). NPCi's AEPS technology gives technological support for this.

10.19 Measures to promote Cashless Economy

- 2019: RBI setup **Nandan Nilekani committee.**
- Budget imposed limits and restrictions on cash transactions (more in Pillar2:Black Money)
- + many small small things for Mains Exam. It's given in Mains Handout. This is Prelims HDT.

10.19.1 Payments Infrastructure Development Fund (PIDF) by RBI

⇒ ₹250cr from RBI + ₹250 cr from (Banks and Card Gateway Operators) ⟶ ₹500 cr fund setup.

⇒ **Objective?** Provide funding/ subsidy to encourage merchants in the small towns, villages and North Eastern states to adopt point-of-sale (PoS) card swiping machines. (भुगतान इंफ्रास्ट्रक्चर के लिए निधि/कोष: गांव, छोटे शहरों और उत्तर-पूर्वी इलाकों में व्यापारियों को कार्ड मशीन अपनाने के लिए मदद)

⇒ 2021: RBI launched it. Presently scheme/fund valid upto Dec, 2025.

Scheme	description
PM SVANidhi	Govt helps the street vendor to get Biz loan. (Ref: Pillar#1D for more)
PM Vishwakarma	Govt helps artisans e.g. carpenter, potter, blacksmith, et cetera. In skill training Bank loans. (Ref: Pillar#1D for more)

RBI said we'll gives PIDF-subsidy to above beneficiaries to adopt card machines.

10.20 🌐🤝 Global Money Transfer

10.20.1 🌐📲 UPI One World (prepaid wallet) for NRI/ foreign tourists in India (2023)

- NRI/ foreign tourists load money in this separate UPI One World App's digital wallet by using their foreign credit/debit cards.
- Then use UPI One World App to scan & pay local Indian vendors.
- Useful because often local small vendors do not have the PoS device to scan foreigner's international credit/debit card.
- 2023: Facility first introduced during the G20 India summit. 2026: India AI Summit 2026- facility given to tourists from 40 nations.

10.20.2 Nostro-Vostro / Rupee-Ruble / Rupee-Rial Agreement? - Ref Pillar3

(Ref: Pillar#3: international trade / currency exchange)

10.20.3 💳 RUPAY Cards- International acceptance

2023-June: RBI allowed international use of RuPay cards. That means an Indian travellers in use them abroad for ATM money withdrawal and shopping. 2024: implemented for Mauritius.

10.20.4 🛍️ UPI in Srilanka, Mauritius etc.

As of 2025-Nepal, Bhutan, Sri Lanka, UAE, Singapore, Mauritius, France, UAE etc allow UPI facility. This helps Indian tourists to pay through mobile phones.

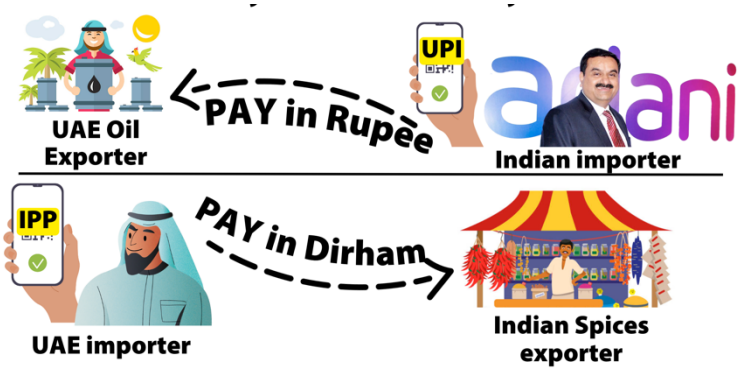
🔤 **MCQ. How many have signed UPI transaction deal with India? (Pre'25)**

I. United Arab Emirates II. France III. Germany IV. Singapore V. Bangladesh

Codes: (a) Only two (b) Only three (c) Only four (d) All the five

Ans. Tough Question. As of 2025-May: answer is - (C) 3 of them - UAE, France, Singapore.

10.20.5 🛍️ UPI@Global: India-UAE: LCSS: UPI-IPP linkage (2023-Aug)



- 2023: Central banks of India and UAE signed Local Currency Settlement System (LCSS) agreement. This will allow importers/exporters/investors/tourists to pay in their respective domestic currencies i.e. in Indian rupee (INR) & UAE Dirham (AED). (Without using Dollars.)
- Both nations also trying UPI-IPP linkage for easier Local Currency Settlement System (LCSS).

Country →	India	UAE
Mobile based Fast Payment Systems (FPSs) →	UPI	Instant Payment Platform (IPP)

📚 More about currency exchange in (Ref: Pillar#3 Currency Exchange)

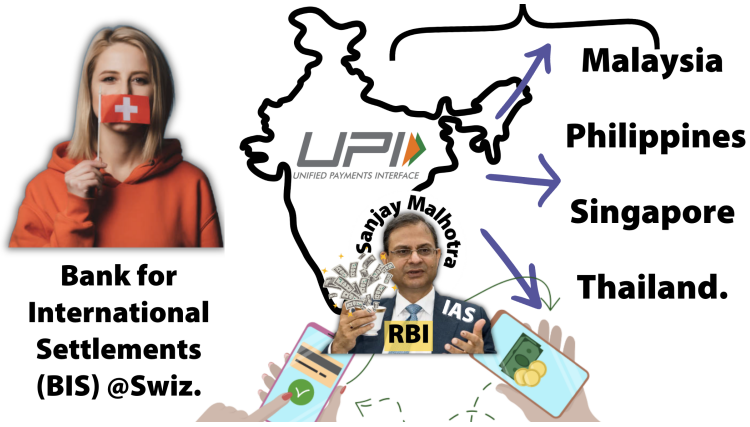
10.20.6 🌍 💳 UAE's JAYWAN card (=RUPAY) & AANI (=UPI) based on NPCi's model

- National Payments Corporation of India (NPCI) is collaborating with UAE's Al Etihad Payments (AEP) for creation of domestic card scheme JAYWAN in UAE.
- UAE's JAYWAN is based on the RuPay card. UAE's AANI is UPI like money transfer tech.
- It'll help UAE to become 'self-reliant' (presently dependent on MasterCard visa, et cetera, foreign players for running the card system)

MCQ. JAYWAN is the domestic Debit Card of which of the following countries? (CAPF-2024)

(a) UAE (b) Egypt (c) Saudi Arabia (d) Nepal

10.20.7 🌍 UPI@global → Project Nexus Indian-UPI connect with some Asian nations (2024)



- **Objective?** To Connect India's Unified Payments Interface (UPI) with the Fast Payments Systems (FPSs) of Malaysia, Philippines, Singapore & Thailand.
- **Nodal/Coordinator?** Bank for International Settlements (BIS, Basel, Switzerland)
- **Benefit?** will help in crossborder payments for Person to Person (P2P) and Person to Merchant (P2M) within 60 seconds.

10.20.8 🛍️ UPI@Global: EU-TIPS linkages (2025-Nov)

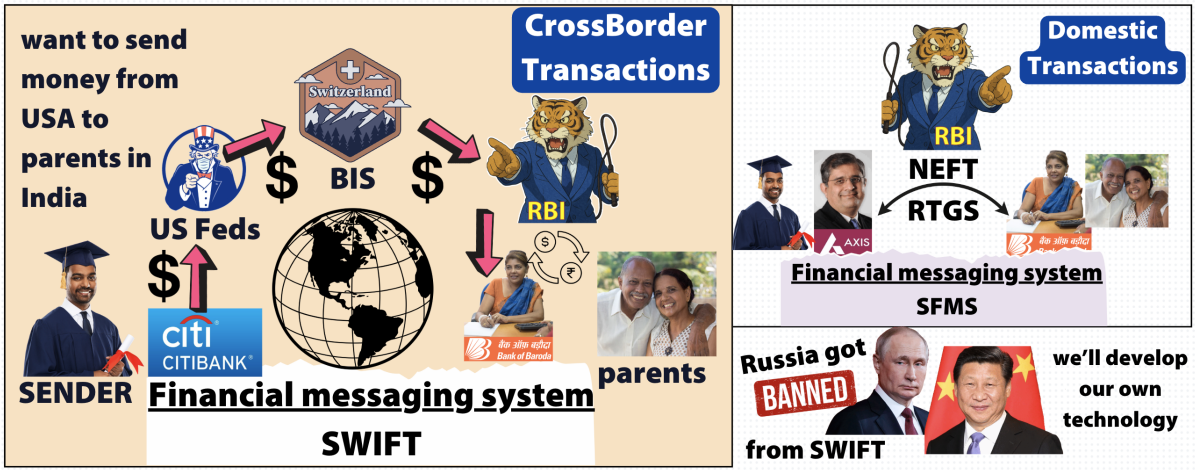
- European Union (EU)'s UPI-like system is called TARGET Instant Payment Settlement (TIPS, Full form, not important). - 2025-Nov: RBI & EU's Central Bank signed deal for UPI-TIPS linkage.

10.20.9 💸📞 Financial Messaging Systems: SWIFT- Russia gets banned

Globally (वैश्विक स्तर पे पैसा भेजना हो)	Within India (भारत के भीतर ही पैसा भेजना हो!)
Society for Worldwide Interbank financial telecommunication **(SWIFT)**	Structured Financial Messaging System **(SFMS)**
By a Cooperative organization in	designed by Tata Consultancy Services (TCS) for

Globally (वैश्विक स्तर पे पैसा भेजना हो)	Within India (भारत के भीतर ही पैसा भेजना हो !)
Belgium's La Hulpe city (1973).	IDRBT (=Research arm of RBI)
To serve the messaging function for Banks, NBFCs & brokers who may / may not have direct bank relations / settlement systems with each other due to international borders.	to serve the messaging function in NEFT, RTGS. **FAQ by Zhande-walle Babushone:** does it also work with UPI/IMPS? Ans. I don't know. I didn't search. Not loudly given in newspapers/books. थोड़ा-पढ़ो-आगे-बढ़ो.

Nations banned from SWIFT: 1) Iran (Capital Tehran) 2) 2022: Russia got ban, after invading Ukraine. This will make it very difficult for the Russian banks/ NBFCs /businessmen to send and receive money internationally (यूक्रेन-आक्रमण के ख़िलाफ़ रूस को स्विफ़्ट-वित्तीय संदेश प्रणाली से प्रतिबंधित किया गया है जिसके चलते रूसी बैंक वित्तीय संस्थान और उद्योगपतियों को अंतरराष्ट्रीय स्तर पर पैसों के लेन में दिक्क़तें आएंगी।)



10.21 Payment System Operators?

Payment and Settlement Systems Act, 2007: RBI gives license to Payment System Operators (PSOs).

Type of PSO	Examples
Umbrella Entity For Retail Payments →	NPCi (learned in previous section)
Card Payment Gateway Operator →	MasterCard, Visa, Rupay
Crossborder / International Remittance	Western Union, MoneyGram
Prepaid Payment Instrument (PPI)	AmazonPay, Mobikwik, Phonepe, PayU, etc
Payment Aggregators (PAs)	They help the merchant to collect payment from customer digitally e.g. Paytm, Amazon, GooglePay, CCBill, CCAvenue, Razerpay, Stripe etc.
ATM Operators	SBI etc's own ATMs White label ATMs by Tata, Muthoot, SREI etc
Financial Market Infrastructure Provider(Ref: HDT-Pillar#1C)	Clearing Corporation of India (CCI): provides digital infrastructure for share/bond/foreign currency deals etc.

10.21.1 New Umbrella Entity (NUE) for Retail Payments System (2019)

⇒ Paytm has rival companies like Mobikwick, Phonepe; Mastercard vs Visa; SBI vs Axis Bank;

⇒ But there is no rival company against NPCi. वर्तमान समय में NPCi का प्रतिस्पर्धी कोई नहीं है

⇒ NPCi is an umbrella entity for retail payments system - it operates in card payment (RuPay), Money transfer (IMPS), Mobile Apps (UPI, BHIM), ATM Network (NFS) etc.

⇒ 2020: RBI issued a proposal, "if any Indian company interested to compete in this (NPCi-like) retail payment segment, we'll license them as new umbrella entity (NUE: खुदरा भुगतान के लिए नई छतरी इकाई). Subject to these technical "XYZ" technical eligibility requirements"

Benefits? More companies like NPCi → more competition → more innovation and cheaper services for users. (स्पर्धा बढ़ेगी तो नए आविष्कार होंगे, डिजिटल भुगतान सेवाएं पहले से भी ज्यादा सस्ती / किफायती/बहेतर हो)

⇒ Facebook, Google, Amazon, Flipkart and others had applied for NUE licences. But none of them met RBI's expectations.

⇒ **2023-Jan**: RBI has put this license process on halt/suspension. (जिन कंपनियों ने अर्जी डाली थी उनके अनुभव और काबिलियत से रिजर्व बैंक संतुष्ट नहीं था. इसलिए अभी रिजर्व बैंक ने इस पूरे मामले को ठंडे बस्ते में डाल दिया है)

10.21.2 DELETING Third Party Application Providers (TPAPs): Google Pay, Whatsapp Pay

Outdated. Controversy was later resolved. So deleting from HDT

10.22 DIGITAL PAYMENT REGULATORY BODIES (नियंत्रण संस्थाएँ)

10.22.1 Digital Transactions Ombudsman (DTO-2019 / लोकप्रहरी)

⇒ RBI designates senior RBI officials at 21 places across India as DTO. They hear customer complaints upto ₹ 20 lakh against prepaid payment instruments, Mobile wallets, Apps, NEFT/RTGS and other digital transactions. (डिजिटल लेनदेन शिकायत निवारण अधिकारी/लोकप्रहरी).

⇒ They can order the company / bank to fix problem and pay upto additional ₹ 1 lakh for mental agony of customer → Higher Appeal to **Dy.Gov of RBI.** (मानसिक उत्पीड़न के लिए जुर्माना/मुआवजा).

⇒ If matter > ₹ 20 lakh, then matter outside his jurisdiction. Victim has to approach courts. (20 लाख से अधिक रकम का मामला होगा, तो पीड़ित ने न्याय/इंसाफ के लिए अदालत/कोर्ट में जाना होगा)

10.22.2 RBI: BPSS (Statutory body) replaced by PRB (2025)

⇒ **1998**: Narsimham-II Committee on Banking Reforms suggested regulatory framework for e-banking, card payment etc. Then,

⇒ **2007**: Payment & Settlement Systems Act → RBI → (Statutory) Board for Regulation and Supervision of Payment and Settlement Systems (BPSS) (भुगतान और निपटान प्रणाली के विनियमन और पर्यवेक्षण के लिए बोर्ड).

⇒ All types of payment system providers have to register with RBI's BPSS- whether bank, non-bank, wallet/Prepaid Payment Instrument (PPI) etc. (सभी भुगतान संस्थाओं ने पंजीकरण करवाना अनिवार्य)

⇒ 2025- this BPSS body replaced by new statutory body called PRB. See next table:

BEFORE	After 2025 reform
Board for Regulation and Supervision of Payment and Settlement System (BPSS)	Payments Regulatory Board (PRB)
Chairman = RBI governor.	SAME
5 members. But, no nominee from the government side. (सरकार की तरफ़ से कोई प्रतिनिधी नहीं)	6 members, including **3 members from the government side** e.g. Secretary (usually IAS) from Electronics ministry, Secretary (usually

BEFORE	After 2025 reform
	IAS) from Dept of Financial Services etc.
How many times they meet? = Not imp.	Must meet minimum two times per year.

10.22.3 RBI: ₹25k/85% Cyberfraud Compensation from Jan'27

- Victim of small online fraud (loss ≤ ₹50,000) gets 85% of loss or ₹25,000, whichever is less.
- **Cost split:** RBI 65% (from Depositor Education & Awareness Fund) + victim's bank 10% + scammer's bank 10% = 85% refund. (मुआवजा मिलेगा).
- If cross-border then RBI 65%, victim bank 20%.
- **Conditions**: Victim must report to bank AND cybercrime helpline (1930) within 5 days. Available once per lifetime per individual. (आजीवन में केवल एकबार मिलेगा)
- Even if victim shared OTP, he'll get this compensation. (भले पीड़ित ने बदमाश को ओटीपी साझा किया हो)
- Note: This is in draft stage. Actual terms and conditions may change in the future.

10.23 Digital Payment related Index / Competitions/Reports by RBI

Below table given if preparing for RBI grade-B officer exam etc. I am not spending time in writing the ranks because poor cost benefit for #IAS-Prelims-RAFTAAR.

REPORT	RBI's Payment and Settlement Systems in India Vision 2019 – 2021
INDEX/Ranking	RBI Digital Payments Index– to check level of digital payment in various areas of India. (DPI डिजिटल भुगतान सूचकांक) (Ref: Pillar#1D3 fin.inclusion for more)
INDEX/Ranking	RBI's Financial Inclusion Index– to check whether every poor person is having access to banking, loan, insurance pension, etc. (वित्तीय समावेशन सूचकांक)
Competition	RBI's Harbinger Hackathon competition for innovation in digital payment.

10.24 Crypto-Currency & Blockchain Technology

Note: For easier visualisation, I have wrote "bitcoin" instead of "crypto-currency" at most places.

⇒ **[Definition] Cryptocurrency:** is a digital / virtual currency created & stored using blockchain technology.क्रिप्टोमुद्रा एक डिजिटल मुद्रा है जिसे ब्लॉकचेन तकनीक का उपयोग करके बनाया और संग्रहीत किया जाता है।

⇒ **[Definition] Blockchain**: A secured decentralized database/ distributed ledger technology (DLT) that maintains a continuously growing list of records / transactions. Old entries can't be deleted, new entries will be visible to all. Mainly used for running cryptocurrency network. But it can also store any type of data. (ब्लॉकचेन एक डेटाबेस है जो रिकॉर्ड की निरंतर बढ़ती सूची को विकेन्द्रीकृत रूप से बनाए रखता है।)

⇒ **[Origin]** Anarchist groups lost faith in **FIAT MONEY** because USA's Subprime Crisis (2007) eroded the purchasing power of US Dollar. They also dislike **Banks & Card Companies** because of transaction charges on e-banking, card payments, MDR, interoperability issues. (अशासनवादी समूहों को सरकार द्वारा जारी की मुद्रा नापसंद। इन अशासनवादी समूहों को बैंक और कार्ड कंपनियों से भी नफरत है)

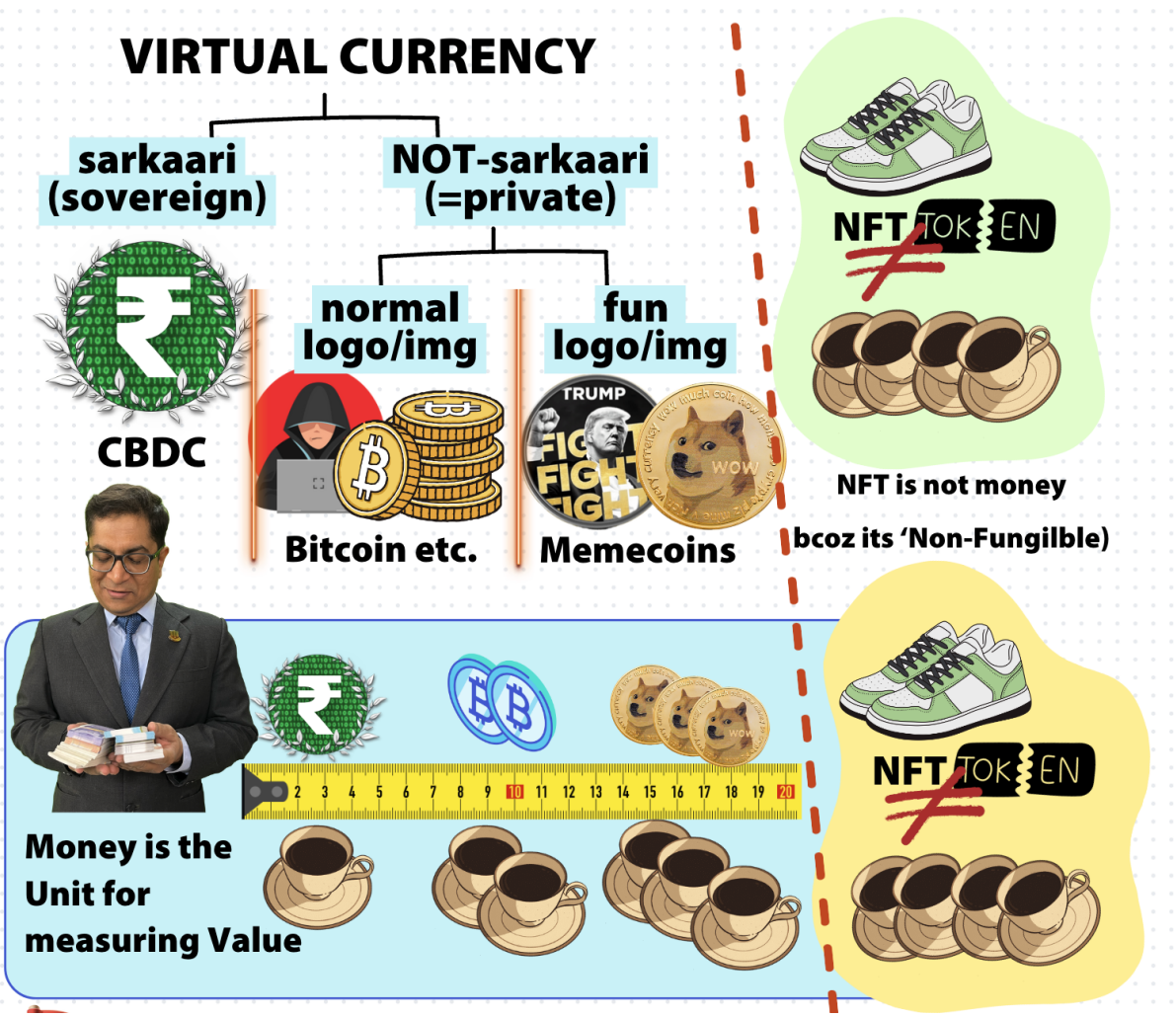
⇒ **2009:** An anonymous user (अनाम/अज्ञात) **Satoshi Nakomoto** launched a cryptocurrency '***Bitcoin***'.

⇒ total 21 million Bitcoins, wherein 1 Bitcoin (BTC) = 10^8 Satoshi (the smallest unit).

⇒ **Other examples:** Ethereum, Litecoin, Digicoin, Laxmicoin, Ripple, Dogecoin etc.

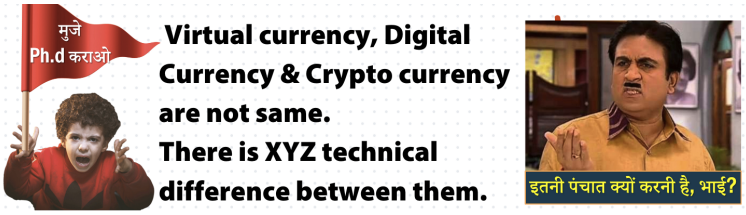
10.24.1 → Virtual currency MEME Coins e.g. Trumpcoin

- A meme coin is a cryptocurrency named after characters, celebrities, animals, artwork, etc to be humorous, light-hearted. not issued by government. Issued by private coders / individuals.



- 2013: Dogecoin (photo of a Japanese dog) is considered to be the first meme coin.
- 2025: $TRUMP meme coin (photo of Trump rising a fist). 1 unit of TRUMP meme coin = $29 as of 25/jan/2025.

Meanwhile, 🚩👶 **Zhande-walle Babushone:**



10.24.2 🥖Hot Wallet vs Cold Wallet

Just like you store money in Paytm DIGITAL Wallet. You can store Bitcoin and other cryptocurrency in Crypto Digital Wallets. Such wallet could be hot wallet or cold wallet.

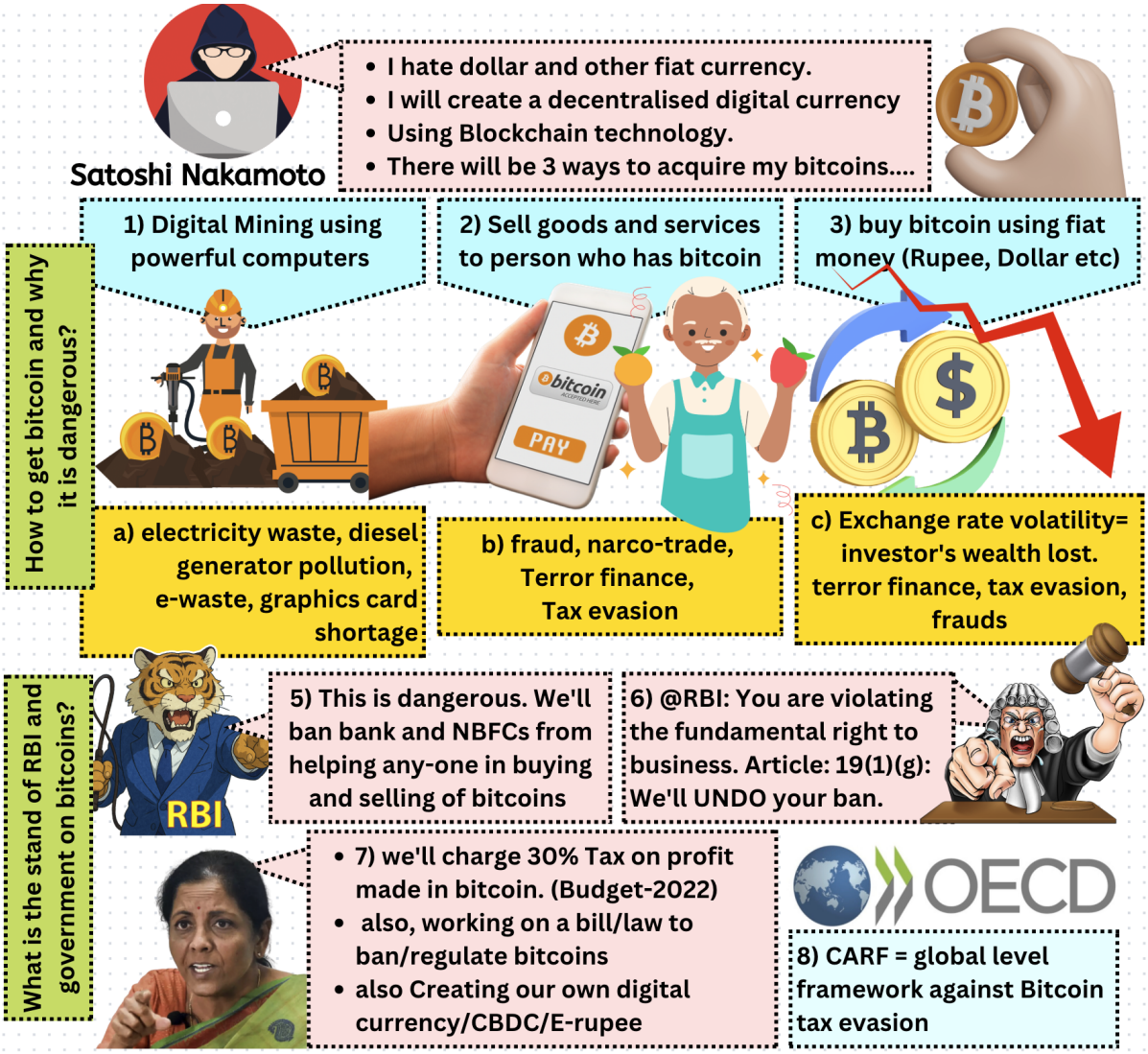
⇒ 🔥 Hot wallets are connected to the internet, (e.g. an online app)

⇒ ❄ Cold wallets are not connected to internet (e.g. a USB pendrive).

10.24.3 🔗🧠📉:🥶☃ Bitcoins prices up and down

2021-22	Bitcoin fell from ~$70,000 to $20,000 ⟶ almost 70% loss! It was called 'Crypto-Winter'

2024-Dec:	1 Bitcoin = $100,000; then in 2025: it crossed above 1 lakh dollars. Bcoz of Pres Trump's policies favoring crypto currencies.
2026-July	Fell to $64,500



10.24.4 → Virtual currency USA Bitcoin Strategic Reserve Fund (2025)

⇒ 1975: U.S. Strategic Petroleum Reserve was created. The USA government purchases oil when it is available at a cheap price. → uses it as a buffer against global supply shocks later.

⇒ Over the years, USA federal officers seized ~200,000 bitcoins from criminals and tax evaders.

⇒ 2025: Trump set up a US bitcoin strategic reserve fund from such bitcoins.

⇒ **Benefit?** (1) Sell Bitcoins at a profit when its price goes up. (2) Use the profit money for welfare programs, etc. (3) Cover the govt's debt without raising taxes.

10.24.5 DELETING Bitcoin Halving = 50% reduction in the reward to miners (2024)

Outdated. Deleting.

10.24.6 🔗🧠📉:🥶🤖 Crypto-scams - Recent Global Scams

Following portals/website/online-exchange sites help users buy/sell crypocurrency.

Binance	Infamous for frauds/scams.
FTX	⇒ Its founder/CEO arrested for scam ⟶ 10 lakh+ investors' money got stuck.
WazirX	⇒ This Indian cryptocurrency portal lost $230+ million in a cyber-attack (2024)
CoinDCX	⇒ $44-million worth bitcoins stolen. (2025)

🚩👶**FAQ:** How were these scams conducted? Ans. Its forensic analysis not important for our exam. Ans. Sufficient to know these are the dangers associated with crypto currency. (**ऑनलाइन प्लेटफॉर्म जहां क्रिप्टो मुद्रा की खरीद बिक्री की जाती थी. 10 लाख से अधिक निवेशकों को चूना लगाया गया/ गबन हुआ है)**

🚩👶**FAQ: How is Bitcoin not traceable, we can trace it using XYZ technology?**

✅Ans. Using HTML, we can create Wikipedia website and using HTML we can also create gmail website. While everyone can read Wikipedia, I can't read your gmail, **unless I know your username password.** Blockchain is similar. It depends on designer to make database "open" or "private/**encrypted**". Beyond that, we are not here for B.Tech (Computer Science) Semester exam/James Bond recruitment. We just have to memorize 3-4 negative points for exam.

10.24.7 🔗🧠:Crypto currency observations by 📙EcoSurvey'23

⇒ Cryptocurrencies are primarily held by a few big players/"whales". So they can manipulate the prices by manipulating the supply.

⇒ Problems related to electricity-consumption, tax-evasion, abnormal fluctuation in prices, investors-fraud, no-intrinsic value et cetera.

⇒ And ideally, we need strict rules over crypto-market, just like we've strict rules over sharemarket

10.24.8 🔗🧠:🍋 Cryptocurrency Profit pe 30% Tax and 1% TDS in 👜🧍 Budget-2022

Refer to Pillar2A: Budget Direct Taxes to learn this matter.

10.25⚖️🔗🧠✋ Cryptocurrency Banned in other nations?

⇒ Nepal, Algeria, Egypt, Morocco, Turkey , Iran etc: Bitcoin is illegal. (कुछ देशों में गैरकानूनी है)

⇒ Chinese prohibited the banks from providing buying/selling/investment services for bitcoin etc cryptocurrencies. After this crackdown, Bitcoin prices fallen ↓↓ by 30%

10.25.1 Crypto currencies developments in some countries

Venezuela	Two legal tenders: 1) Bolivar Paper currency issued by Govt 2) Petro Cryptocurrency issued by Govt.
Marshall Islands	**Capital:** Majuro, Sovereign state having free association with USA (for defense, social security, budget). Legal Tender was U$D. But, ⇒ 2018-Feb: became **first country** to launch sovereign cryptocurrency named "Sovereign" [SOV]. So, now two legal tenders: US Dollars and SOV.
El-Salvador	⇒ El-Salvador: Two Legal tenders [1) US Dollars. [2) Bitcoin (from 2021). ⇒ Thus, El-Salvador = 1st nation in the world to allow Bitcoin as legal tender
Bhutan	• Bhutan requires foreign tourist to pay daily tourist fee (in addition to Hotel,

	food, et cetera,) but 2020: Corona → Tourists flow ⇓. • 2023: Bhutan decreased tourist fee from $200 to $100 to attract tourists • 2025: Bhutan allows tourist to pay for hotels, food, taxi etc using cryptocurrency, with help of a Crypto-company named Binance.
Pakistan 2025	⇒ Pakistani public unable to afford the electricity bills so Pak. Govt allotted 2,000 megawatts (MW) of electricity for cryptocurrency. (with hopes of attracting foreign investors / miners in Pak.) ⇒ 2025: official Pakistan Crypto Council (PCC) signed deal with Trump's company World Liberty Financial (WLF) for bringing blockchain innovation, stablecoin adoption etc. in Pakistan

10.25.2 🔗🧠:🤩 Cryptocurrency Permitted in Other Nations/Groups

🌐 **UNICEF**	⇒ United Nations International Children's Emergency Fund (UNICEF, HQ-New York,USA; 1946, later renamed to UN Children's Fund) ⇒ 2019: UNICEF setup a Cryptocurrency Fund to accept donations in cryptocurrencies. It has become the **first** UN organization to accept cryptocurrency. (यूनिसेफ संस्था दान में क्रिप्टो मुद्रा का भी स्वीकार करती है)
🌐 **World Bank (More about this org in 🗒️ Pillar#3B)**	⇒ 2018: World Bank launched World's **first** blockchain bond called "Bond-i" in Australia, **denomination: Australian Dollars** → public invests, gets ~2% interest after 2 years. ⇒ Blockchain bond requires less cost in server / database /paperwork maintenance unlike traditional bonds. 🗒️*More in Pillar#1C: Sharemarket*

10.26🔗🧠🥇 Cryptocurrency → Global Regulations

10.26.1 Global Regulations → MICA regulation by EU (2023)

⇒ European union (EU)'s rules called the Markets in Crypto Assets (MiCA)

⇒ To protect people from scams /frauds in crypto currency investment.

⇒ To combat tax-evasion, money laundering (i.e. converting black money into white) AND terror-finance. Doesn't regulate central bank digital currencies (CBDC).

⇒ Doesn't regulate nonfungible tokens (NFTs).

10.26.2 Global Regulations → CARF Framework by OECD (2022)

⇒ Crypto-Asset Reporting Framework By OECD (Organisation for Economic Co-operation and Development) Objective? Similar to EU's MICA Rules.

10.26.3 🔗🧠🥇 Cryptocurrency → Stable-coins & Genius Act

⇒ Stablecoin is a type of cryptocurrency whose price is backed by a reserve asset like gold / dollar / fiat currency e.g. Facebook's proposed 'Libra', Tether, Basecoin, and TrueUSD.

⇒ 😨But, stablecoins challenges: 1) money laundering (धनशोधन / काले धन को वैध बनाना) 2) terror financing (आतंकी वित्तपोषण) 3) lack of consumer protection (उपभोक्ता संरक्षण की कमी) 4) financial frauds (वित्तीय धोखाधड़ी). **2019**: France, EU, G20 group opposed. (बड़े देश/संगठन इसके विरोध में है)

⇒ **2025**: USA Prez Trump brings GENIUS Act to regulate Stable-coins in USA.

10.26.4 ES26 worried about growth of Stablecoins

⇒ 2025: Stablecoins market rose to 300+ billion$.

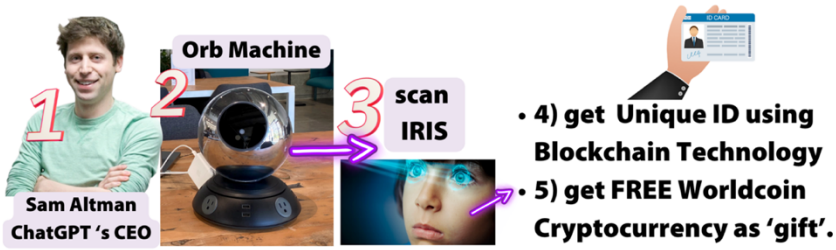
⇒ Stablecoins' prices are linked to traditional physical and financial assets.

⇒ It can cause negative spillover to economy.

10.26.5 USA 3 laws on cryptocurrencies

Act/Bill	Purpose
CLARITY Act	To Regulate Cryptocurrencies
GENIUS Act	To Regulate Stablecoins
Anti-CBDC Surveillance State Act	To prevent USA's Central Bank (US Federal Reserve) from misusing CBDC/digital currency for surveillance of citizens.

10.26.6 World Coin Project by ChatGPT's Sam Altman (2021–22)



Pros and Cons? Similar to the Aadhaar card:

- Pros: IF governments across the world ACCEPT this as "identification proof"= easier to avail various government schemes benefits /college admission / passport application etc. Time-trouble saved. (यदि देश विदेश की सरकारें इसे एक आधिकारिक पहचान पत्र के रूप में मान लें तो गरीबों को विविध योजना स्कूल कॉलेज दाखिले आदि में काफी सुविधा हो सकती है) Cons: concerns over privacy. (निजता का हनन. दुरुपयोग का भय)

MCQ. Find correct statement(s) about Bitcoin? [UPSC-CDS-2017-I]

(1) It is a decentralized virtual currency. (2) It is generated via complex computer software systems.
(2) Reserve Bank of India has recognized Bitcoin as a legal tender in January 2016.
Answer Codes: (a) 1 only (b) 1 and 2 only (c) 2 and 3 only (d) 1, 2 and 3

MCQ. Find correct statement(s) about "Blockchain Technology" [UPSC-Prelims-2020]

1. It is a public ledger that everyone can inspect, but which no single user controls.
2. Structure and design of blockchain is such that all the data in it are about cryptocurrency only.
3. Applications that use basic features of blockchain can be developed without anybody's permission. Answer Codes: [a) 1 only [b) 1 and 2 only [c) 2 only [d) 1 and 3 only

10.27 Central Bank Digital Currency (CBDC) केंद्रीय बैंक डिजिटल मुद्रा

Year	Nation	Paper Currency	CBDC
2020	Bahamas (Capital Nassau) – 1st nation in the world to issue CBDC##	Bahamaian Dollar BSD	"digital B$", also called 'sand dollar'. issued Non-Interest Bearing CBDC
2021	Nigeria (Capital Abuja)	Naira	the "eNaira"-Non-Interest

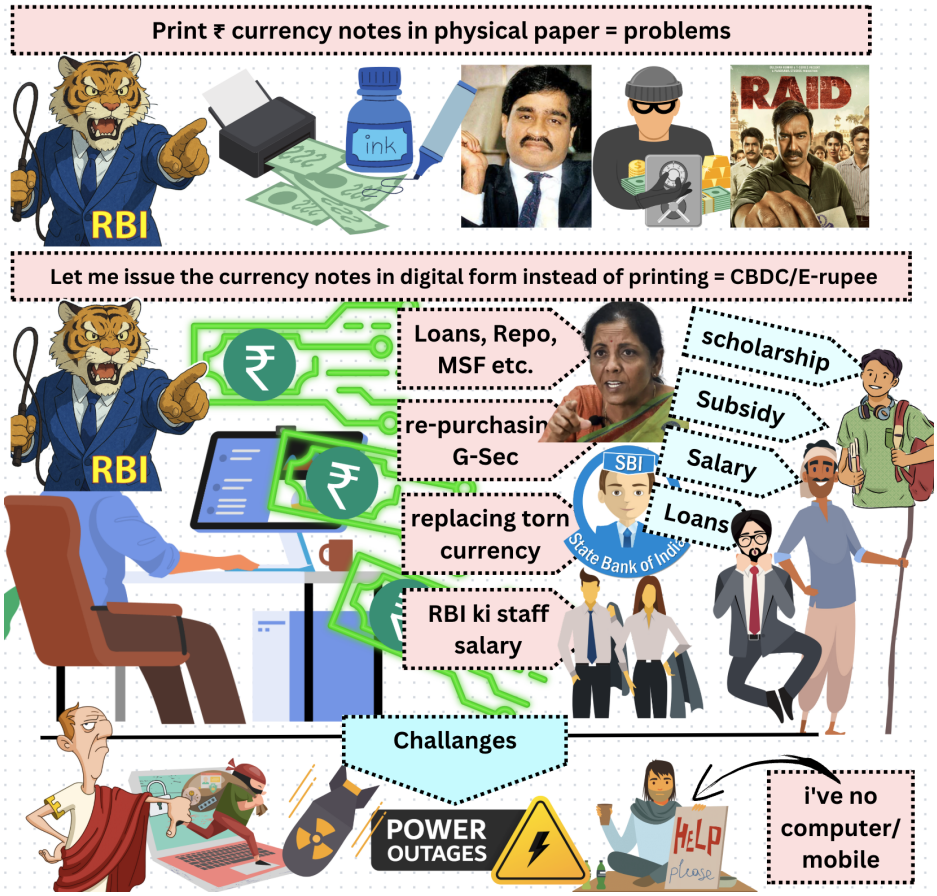
Year	Nation	Paper Currency	CBDC
			Bearing CBDC.

##Marshal island launched SOV currency in 2018 then shouldn't we call it the first country to launch CBDC? why above table says Bahamas is 1st country to launch CBDC?

⇒ Ans. Big Newspapers using such headlines. I didn't bother to write/ask to their chief-editors. If asked in Exam, then tick MCQ depending on options.

⇒ China, Britain, Japan, Singapore, Sweden have also started such experiments.

⇒ Budget-2022 announced issuance of Digital Rupee using blockchain technology).



10.27.1 Zhande-walle Babushono ke liye special instruction: (अग्रिम सूचना)

- Following are the features told by the (A) Govt to the Parliament and (B) RBI governor in a press release. So, accept and memorize it.
- Please **do not waste your time engaging in academic/legal debates - if you value your time and wish to complete the syllabus in a timely fashion.**

10.27.2 Indian CBDC symbol (e₹)

Normal rupee = ₹ or INR. CBDC rupee = e₹ or eINR.

10.27.3 (e₹) CBDC wholesale vs Retail

⇒ Presently RBI doing trials/experiment/pilot-studies in two segments:

Type	CBDC-Retail (e₹-R)	CBDC-Wholesale (e₹-W)
Trial-Purpose	• Person to Person (P2P) • Person to Merchant (P2M) transactions.	• Secondary market transactions in government securities • Interbank borrowing for short-term loans • Crossborder payments.
Details	given in present handout 1A1	Ref: Pillar#1C to understand these terms.

10.27.4 (e₹) Indian CBDC = digital token of legal tender

⇒ Indian CBDC (e₹) is based on blockchain technology.

⇒ Indian CBDC (e₹) is a digital token that represents legal tender.

⇒ (e₹) created under RBI Act, 1934 by amending some sections in 2022.

⇒ RBI has issued (e₹) on behalf of Government of India. [FAQ: **I have legal arguments against this.** Ans. ACCEPT it- bcoz Government said this in Parliament.]

⇒ (e₹) is a **liability** of the RBI e.g. similar to cash note " I promise to pay the bearer on demand sum of ₹100 rupee".

⇒ It is being distributed through financial intermediaries, i.e., banks.

⇒ Just like cash, it (e₹) will not earn any savings interest.

⇒ It offers features of physical cash like trust, safety and settlement finality.

MCQ. Find WRONG about UPI vs CBDC? (IAS-2026)

(a) UPI is a real-time payment system but Digital Rupee is akin to sovereign paper currency.

(b) In case of UPI, settlement for end users happens instantly as the money gets immediately debited or credited but in case of Digital Rupee, there is no settlement as the wallet balance gets transferred to another wallet.

(c) UPI transactions are recorded by banks and reflected in bank statements but in case of Digital Rupee, no data is captured in bank statements as transactions are from one wallet to another.

(d) ✗ In both the cases (UPI and Digital Rupee), the liability lies with the users and their respective banks. Hint: e₹ is liability of RBI. (I promise to pay bearer….theory) & we've to find wrong st.

10.27.5 (e₹) → Settlement finality

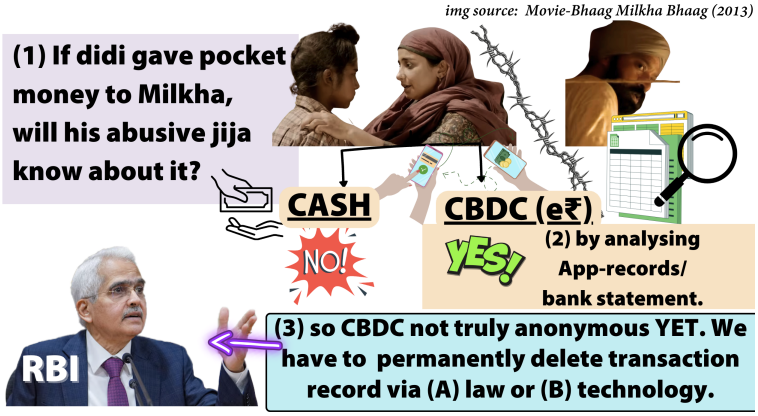
⇒ "settlement finality" = there is no risk of reversal or cancellation after the financial transaction is complete. e.g. Cash or CBDC-e₹. (एक बार भुगतान कर दिया तो फिर सामने पार्टी के पास पैसा गया। उसको उलट नहीं सकते।)

⇒ Whereas, if a payment is made via Credit card/ cheque-book → customers can later block the payment by contacting the banker within a time-limit. So they (creditcard / cheque) do not have instant-settlement finality. (उधर आप बैंक में शिकायत कर के भुगतान को रोक सकते हैं।)

⇒ **FAQ: can we say that as mode of payment (UPI/Debit Card/NEFT-RTGS) also has settlement finality?** Ans. I copied only that much which is given in Newspaper/Press-release. I did not become Sherlock Holmes to do find further info/PhD. Feel free to delay your syllabus completion by asking ChatGPT.

10.27.6 → CBDC Permanent deletion of transactions to keep e₹ anonymous like cash

img source: Movie-Bhaag Milkha Bhaag (2013)



Note: This is proposed. This is not yet done. As of 2026. सीबीडीसी द्वारा भुगतान के वक़्त नक़दी जैसी गोपनीयता नहीं रहती। इसलिए जबतक सीबीडीसी के लेनदेन के डिजिटल रिकॉर्ड को क़ायमीरूप से डिलीट न किया जाए, इसमें १००% गोपनीयता नहीं है।

FAQ by Zhande-walle Babushone: above graphics contradicts with the MCQ asked in 2026 about UPI vs CBDC (option-C-UPI transactions are recorded by banks and reflected in bank statements but in case of Digital Rupee, no data is captured in bank statements as transactions are from one wallet to another.) **Mrunal**- RBI governor is talking about the future challenges when e₹ testing-experiment is no longer bound/confined within 'wallet'.

10.27.7 (e₹) Indian CBDC = no interest earning (नहीं मिलेगा ब्याज।)

- (e₹) stored in a separate digital wallet/separate mobile created by the bankers.
- (e₹) does not earn savings interest rate, unlike normal bank deposits (because you are keeping it with yourself in a digital wallet. So bankers do not treat it as a deposit).
- (e₹) can be converted to other forms of money, like deposits with banks. But then you will

have to move that money out of your digital wallet → then you cannot use it as (e₹) [ब्याज कमाना है तो उसको डिजिटल बटुवे से स्थानान्तरित करना होगा।]

10.27.8 🐼 China to pay interest on Yuan CBDC/e-CNY (2026)

- Originally, Chinese Yuan CBDC (labelled 'e-CNY') was non-interest bearing in nature.
- 2026-Jan onwards: e-CNY wallet balances will earn interest rates for the user similar to the normal savings deposit made in banks.
- Further, PhD, not important at this stage until either RBI or the Economic Survey makes more observations.

10.27.9 💮(e₹) CBDC - programability & offline features

- Presently (e₹) can be used for Person to Person (P2P) and Person to Merchant (P2M) transactions. 2024-Feb: RBI proposed two new experiments:
- (1) Offline functionality in areas with poor or limited internet connectivity.
- (2) Programability. (given below) 👇

Programability	Example
Biz Travel	⇒ e.g. If Adani allows a company budget of ₹10,000 (e₹) for his employees to use for buying railway tickets only. → then employees cannot use it for buying table chairs, IPL tickets, etc.
Expiry/Validity	⇒ e.g. The university gives ₹10,000 (e₹) stipend to students to buy books within 1 month of admission. ⇒ If student does not buy the books, then the money will be returned back to the university.
Geographical	⇒ e.g. It cannot be used outside these XYZ GPS coordinates.

👩‍🦰FAQ: Aren't these features similar to the e-Rupi gift cards by NPCi?

Ans. Don't make khichdi of RBI's product vs NPCi's product. View them as separate photo frames. Observe the PYQs. UPSC unlikely to ask you compare/contrast between them two.

🐂✅🐂✋ Be focused on the track like a race horse. Don't loiter like a street-bull.

10.27.10 E-Rupee vs E-Rupi

Term	E-Rupee	E-Rupi
Founder	RBI	NPCi
Feature	Central Bank Digital Currency.	Digital GIFT CARD / SMS Code to help beneficiary do shopping.
Fiat money and legal tender?	Yes	No. Because, this is not money. This is a tool to use money.

10.27.11 🐯🔗💮=👌🥰 Benefits of CBDC (केंद्रीय बैंक डिजिटल मुद्रा के फायदे)

CBDC/ Sovereign Digital Fiat Money / Digital Base Money will provide following benefit

1. ↓ Cost **printing and transporting** physical currency. (नोट छापने की लागत और परिवहन खर्च में कमी)
2. ↓ Scope for **counterfeiting**. (जाली नोट बनाना मुश्किल)

3. ↑ **Traceability, ↓Anonymity:** (ढूँढना आसान, गुमनाम/अज्ञात रहेना मुश्किल)
    a. Whether money is reaching the intended beneficiary or not? (लाभार्थी को योजना का पैसा मिला कि नहीं)
    b. From where did the businessman accumulate this money, did he pay an appropriate amount of tax or not? Else auto-alert to the Income tax department for tax evasion. (करचोरी पर निगरानी आसान)
4. Possible to develop programmable currency. E.g. govt transferring ₹6000 in poor farmer's account with condition ke it must be spent WITHIN next 15 days for buying seeds/farming tools ONLY.
5. Global money transfer more easier. E.g. in future China may give DIGITAL YUAN currency LOANS directly to Ghana Without converting into US Dollars and without using SWIFT messaging system.

+ many other things but they are more relevant for Mains rather than Prelims.

10.27.12 🐯🔗💮=✋🥵 Challenges of CBDC (चुनौतियां)

1. 😰 Digital Divide, Financial Literacy, Access Points: (डिजिटल उपकरणों की उपलब्धता की खाई, वित्तीय साक्षरता का भाव, डिजिटल भुगतान इस्तेमाल करने के पहुंच बिंदुओं की कमी)
    a. Not everyone has electronic gadgets or internet connectivity, (सबके पास इंटरनेट नहि)
    b. Not everyone knows how to operate bank account digitally, (सबको इस्तेमाल करना नहि आता)
    c. Not every village shopkeeper accepts payments in the digital form. (दुकानदार नक़दी माँगते है)
2. 😰 Cyber Terrorism/Nuclear Attack/Natural Disaster. If electricity stops → economy will stop functioning. (साइबर आतंकवाद / परमाणु हमला / प्राकृतिक आपदा: विद्युत उपकरण बंद तो अर्थव्यवस्था बंद)
3. 😰 Maintaining the blockchain → 24/7 use of server/electricity → e-waste, pollution. (प्रदूषण)

10.27.13 💮📲 (e₹) CBDC - Interoperability permitted with UPI QR Codes

- 2022–23: CBDC trials/experiments started in a few cities.
- So bankers have developed separate apps to use CBDC money

App	Before	AFTER 2023-Sept
(1) SBI Yono App (Normal money)	can scan & pay UPI-QR codes	can scan & pay UPI-QR codes
(2) SBI-e₹ App (To use CBDC money)	can't scan UPI-QR codes. It required separate codes	can scan & pay UPI-QR codes
Interoperability?	✋🚨Absent/Denied by RBI	✅Present / permitted by RBI

10.27.14 💮📲 (e₹) CBDC wallets: Non-banks can also create wallets (2024-April)

Before	After
banks had created wallets/apps for using CBDC e.g. (1) HDFC Bank Digital Rupee App (2) eRupee App by SBI	yes available
Non-banks e.g. Phonepe, Amazonpay **could not** create such wallet/apps.	Now RBI allowed

🔠 **MCQ.Pre-2023: Find correct about Central Bank digital currencies,:**

(1) It is possible to make payments in a digital currency without using US dollar or SWIFT system.

(2) A digital currency can be distributed with a condition programmed into it such as time-frame for spending it. **Codes**: (a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2.

🔠 **MCQ.Pre-2024: Find correct about digital rupee:**

1. It is a sovereign currency issued by the RBI in alignment with its monetary policy.
2. It appears as a liability in RBI's balance sheet. 3. It is insured against inflation by its very design.
4. It is freely convertible against commercial bank money and cash.

Codes: (a) 1 and 2 only (b) 1 and 3 only (c) 2 and 4 only (d) 1, 2 and 4

Note: we'll solve this MCQ after finishing Pillar1A2 (monetary policy) since terms given in st1 and 2 required understanding of Pillar#1A2.

10.28 💠≠✳️ FUNGIBILITY?

Item	Can it be subdivided in smaller units?	Is mutual substitution possible in terms of value?	Is it fungible?
₹500 Note	Yes ₹100x 5 Notes=500	Yes ₹100 x 5 Notes can buy same quantity of goods as a single ₹500 Note	✅ Yes
1kg gold bar	Yes 100gm x 10 biscuits=1kg gold bar	Yes 100gm x 10 biscuits can buy same quantity of goods as a single 1kg gold bar	✅ Yes
1Bitcoin	1BTC=10^8 Satoshi coins	Yes 1BTC's value = 10^8 Satoshi coins. So, theoretically, they buy same quantity of goods	✅ Yes
100 gm single Diamond	Bigger single piece diamond considered to be more precious/valuable than small-small pieces diamond. So, 100gm Single_Big_Diamond =NOT equal 1gm x 100 small_Diamonds, in terms of value		✋No
Shares, Bonds	NOT possible (How Ref: Pillar#1C: SEBI sharemarket) 🚩👶 FAQ by Zhande-walle Babushone: We disagree, we've counter arguments. Mrunal- I've copied from textbooks. तो कबुल करो।		✋No
NFT	NOT possible. Explained in separate section.		✋No

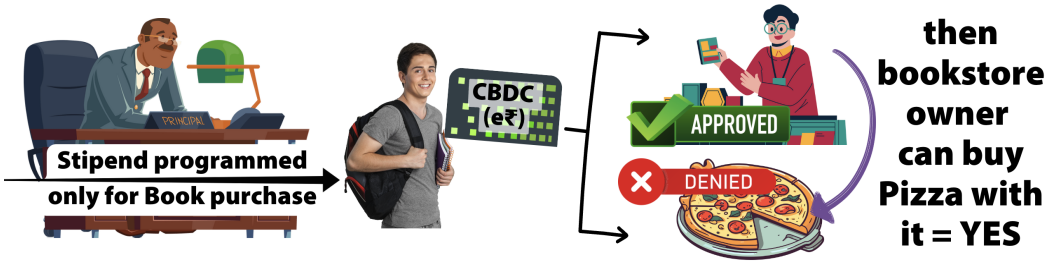
10.28.1 ₹(e₹): Fungibility of CBDC

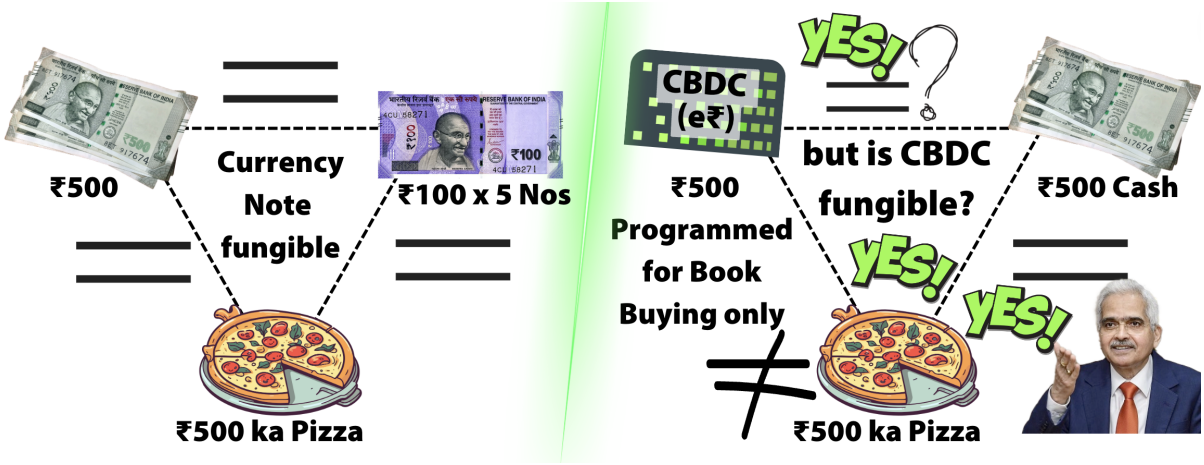
Doubt: if we program the CBDC currency for specific use - example only for purchasing of books, then it is no longer fungible! Because we cannot use it for any other purpose e.g. buying Pizza!!

⇒ In that case, as soon as the book is purchased, the money goes to the bookshop owner → money again becomes fungible. Because the bookshop owner can use the money for any purpose

⇒ ***If the student doesn't buy the book, → CBDC-money goes back to the school, and fungibility is***

restored (because the school admin can use it for any other purpose).





⇒ In that case, as soon as the book is purchased, the money goes to the bookshop owner → money again becomes fungible. Because the bookshop owner can use the money for any purpose

⇒ If the student doesn't buy the book, → CBDC-money goes back to the school, and fungibility is restored (because the school admin can use it for any other purpose).

***"Thus, programmability does not violate/destroy fungibility (of Rupee), it is only a specific use binding."*- said RBI Governor. (भले को शर्ती-प्रोग्रामिंग से CBDC ऐसा सिमित कर दिया जाए, की कुछ ही चीजे उससे खरीद सकते है. लेकिन फिर भी CBDC-रुपया फंजीबल है.)**

10.28.2 💠 Non-fungible token (NFT: ग़ैर-फ़ंजीबल टोकन)

⇒ (Definition) NFT is a Digital file, photo file (JPEG), Animated image (GIF), music file (MP3) etc Stored using blockchain Technology. ब्लॉकचेन प्रौद्योगिकी द्वारा संग्रहित फ़ोटो, संगीत इत्यादि की डिजिटल फाइल है।

⇒ NFT may also be used in storing physical assets. E.g. Ownership document of house, boat, physical-painting etc. (भौतिक सम्पत्तियों की मालिकी दस्तावेजों का भी डिजिटलकरण कर एनएफ़टी में संग्रहित कर सकते है)

⇒ NFT cannot be subdivided. Their individual sub-units cannot be exchanged with one another. Because their values are different based on buyer's preference. So NFTs are non-fungible. (अलग अलग चित्र/संगीत कृतिओं का मूल्य, ग्राहक की पसंद से हिसाब से अलग अलग हो सकता है इसलिए उन्हे छोटे टुकड़ों में विभाजित करना / आपस में उनकी अदला-बदली नहीं हो सकती.)

⇒ Mass Hysteria going on to purchase such digital assets from new-age artists using (mostly) Ethereum cryptocurrency. (नए ज़माने के डिजिटल चित्र/धुन के एनएफटी ख़रीदने का सामूहिक पागलपन चल रहा है.)

⇒ e.g. American digital artist Beeple's Painting JPEG auctioned for $70 million

⇒ E.g. Robot Sophia created a digital painting/art "Sophia Instantiation" (NFT)→ Auctioned for more than 6 lakh dollar$.

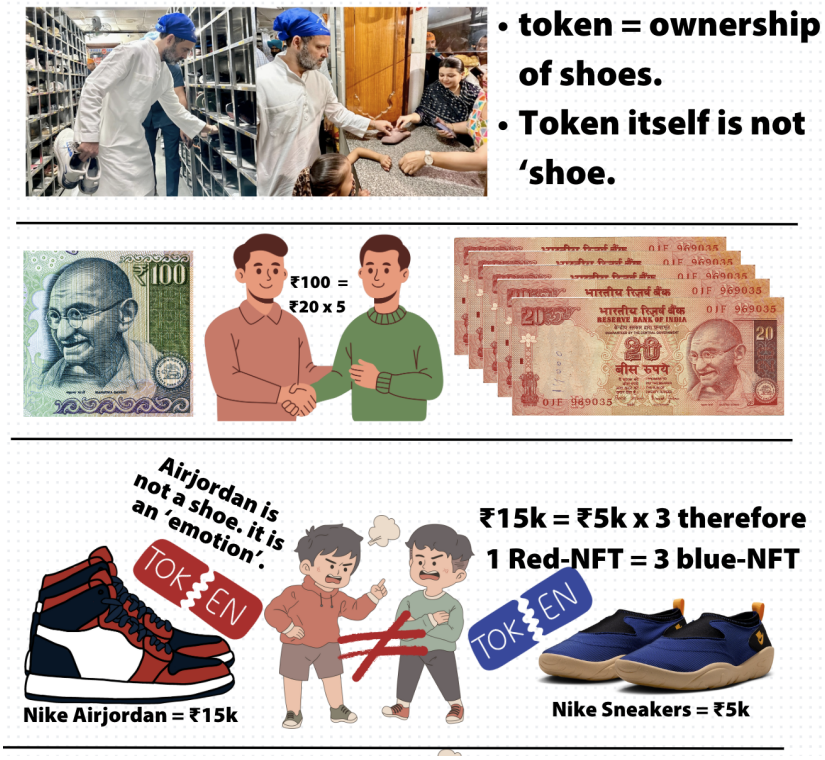
⇒ Many sportsmen, filmstars also creating NFT out of their posters/autographs but we'll not waste preparing all that General Knowledge (GK). (मशहूर खिलाड़ी और फ़िल्म अभिनेता भी इसे जारी कर रहे हैं)

10.28.3 💠😂NFT-Charms? (आकर्षण)

⇒ 1) Multimedia Creators: Easier to trade digital assets (डिजिटल चित्रकार/धुनकार के लिए कृतियों की बिक्री)

⇒ 2) Easier to verify ownership records. Prevent plagiarism. (संपती की मालिकी/स्वामित्व की जाँच करना आसान. नकलचियों के लिए जालसाजी करना / बनावटी दस्तावेज बनाना मुश्किल.)

⇒ 3) Cinema / Sports tournament / Travel tickets etc can be stored with ownership records. Helps in Paperless administration. (सिनेमा / खेल /परिवहन की टिकट की बिक्री डिजिटल स्वरूप से. काग़ज़ मुक्त प्रशासन.)



10.28.4 🔗 Types of Blockchain & National Blockchain Framework (NBF) — MeitY

- **Boss?** MeitY (Ministry of Electronics and Information Technology)
- **Types of Blockchain :**

Type	Access	Hypothetical Examples
Public	Anyone can join, verify, and add data-blocks	Bitcoin and other Crypto currencies
Private	Only selected participants — controlling entity sets rules	Govt applications e.g. Court sending bail/prisoner release order to the Jail superintendent.
Consortium	Semi-decentralized — governed jointly by multiple orgs	e.g. Edu Min & Skill Ministry share database for students' training.
Hybrid	Mix of public + private block-chain	not imp. / complex case.

🔠 **MCQ. Find correct about blockchain technology? (IAS-2026)**

1. Database may be made visible to relevant stakeholders without risk of alteration.
2. Copies of the entire database are stored on multiple computers on a network.
3. Consortium blockchain is a blend of public and private blockchains allowing selective data access.

4. Mathematical algorithms make it impossible to change or delete any data once recorded and accepted. Code: (a) 1 and 3 (b) 2 and 4 only (c)✅ 1, 2 and 4 (d) 1 and 4 only

Hint: 2 and 4 are correct so we're 50:50 between B and C.

10.28.5 🎰 Tokenisation for Financial Inclusion?

- Financial inclusion = improving access to banking, credit (loans), insurance, and pension services to the poor / weaker section. (वित्तीय समोवेशन- गरीबों की बैंक नेटवर्क तक पहुँच बढ़ाना)
- Suppose a small shopkeeper creates his physical asset's NFT token (e.g. property papers or vehicle ownership linked with Aadhar card) → pledges the NFT-token on a PayTM-like app to borrow money without making a physical trip to the bank. This is not done yet in real life. But thus, theoretically, tokenization can help in financial inclusion— as per the UPSC Prelims-2026 official answer key.

🔠 **MCQ. Find correct about Real-World Assets (RWA) Tokenization? (IAS-2026)**

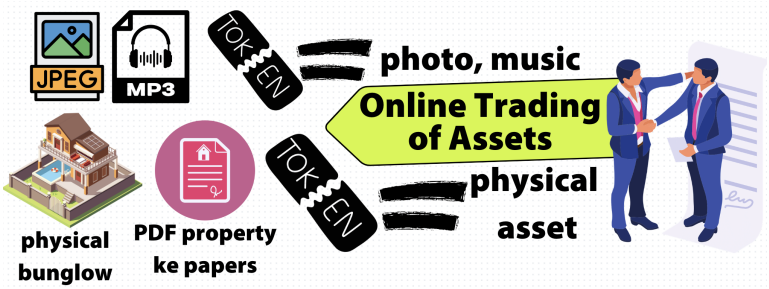
1. Tokenization is the process of turning real world assets into digital tokens using blockchain.
2. Tokenization of real world assets offers 24x7 access, promoting financial inclusion.
3. Tokenization of real world assets will allow the access to high growth investment opportunities for individuals in India. Codes: (a) ✅ 1, 2 and 3 (b) 2 and 3 only (c) 1 and 2 only (d) 1 and 3 only

10.28.6 💸😨NFT-Challenges: (चुनौतियां)

⇒ 1) Started in 2017→ steep price rise in 2021. Single GIF /meme file selling for millions dollar$ worth of cryptocurrencies. This mass-hysteria / investment-bubble will collapse. Investors will lose money. (ये सामूहिक पागलपन /निवेश का गुब्बारा फट जाएगा तब निवेशकों का पैसा डूब जाएगा.)

⇒ 2) Money laundering & terror Finance. How: Ref Pillar#2. (धनशोधन और आतंकी वित्तपोषण में दुरुपयोग)

⇒ 3) Maintaining such computer blockchain record → Electricity consumption ↑→ CO_2 Emission. (इस डेटाबेस का संचालन में बहूत ज़्यादा बिजली खपत होती है, जिससे कार्बन डाइऑक्साइड उत्सर्जन में बढ़ोतरी.)



🔠❓ **MCQ. Find correct statements abt Non-Fungible Tokens (NFTs) (Prelims-2022)**

1. They enable the digital representation of physical assets.
2. They are unique cryptographic tokens that exist on a blockchain.
3. They can be traded or exchanged at equivalency and therefore can be used as a medium of commercial transactions. Codes: (a) 1 & 2 only (b) 2 and 3 only (c) 1 and 3 only (d) 1, 2 & 3

10.28.7 🤷 Epilogue- Babushone want only concepts not facts

🤷 **FAQ: This handout contains so many facts. Can I skip it and just focus on the conceptual part?**

Ans. Just for people like you, I have embedded PYQs inside this handout. Plz observe & reflect.

📄**Next Handout**: Pillar1A2: Monetary Policy: CRR, SLR, Repo etc.